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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgement

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Abstract

The title of this project is Ghar Setu; it is a Django based web app used as an online web-based home rental application that aims at fulfilling the needs of property owners and tenants in the domestic rental market. The system is named Ghar Setu or Home Bridge in Nepali, which shows the essence of the system since it is created to connect people seeking to rent houses with those who can offer their houses, all in one digital platform.

The current rental mechanism used in the target area is highly based on the word of mouth, physical location boards, and random personal communication which is time consuming, geographically restricted and is also liable to miscommunication. Ghar Setu will solve these issues by offering an extensive web application through which property owners can create and manage detailed rental listings complete with images, pricing, property type, contact information, and exact map-based location selection with the help of Leaflet and OpenStreetMap. All available listings are available to the tenants to browse, before filtering them by location and price or viewing them in full property details and can save them to a personal favorites list to be referred to later.

The booking process in Ghar Setu is entirely carried out online. The tenants are allowed to send booking requests to the property owners whose dashboard will receive instant **notification,** and they can accept or deny the request. After the acceptance, a real-time chat room will be automatically opened between the owner and the tenant with Django Channels and **WebSocket**, so that they can communicate directly without using any external messaging platform. Star ratings and written reviews can be left by users on both sides, which encourages transparency and trust within the community on the platform. The system has been built based on Django MVT architecture, MySQL as the relation database, Bootstrap as the responsive web-based front-end design and JavaScript to facilitate dynamic client-side interactions. **The result** is a fully operational, feature-rich and locally focused home renting site that can help in streamlining the entire process of rental experience, as far as property search and booking confirmation and further communication.

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1 Introduction

This chapter presents Ghar Setu and gives the necessary background under which the system was developed and conceived. It provides a history of the issue of home rentals in the target area, defines the key stakeholders and customers who will be positively impacted by the digital solution, and outlines the exact practical situation that led to designing this digital solution. At the conclusion of this chapter, the reader will understand what exactly Ghar Setu is, what it is targeted at, and why it is timely and important to develop it.

1.1 Introduction to project

Ghar Setu is a web-based home rental management system that was created based on the Django framework. It falls under the property technology category, also known as Protech intended to support the local needs of the Nepali market in the field of renting housing. Ghar Setu literally means Home Bridge in Nepali, which reflects the purpose of the system to bridge the gap between property owners and potential tenants with the help of the convenient and trustworthy online platform.

This system has two major user roles. The owners are those who are owners, in other words people or the agency who owns residential or commercial property and would like to advertise the same to a broader population. One of the target audiences is individuals or families actively seeking rental accommodation. The two groups of users access the system via a common web interface, and permissions and role-specific dashboards are used to restrict what each actor is allowed to view and do.

Ghar Setu incorporates an entire system of features in one platform. Owners can register, make elaborate property listing with photos, specify rental charges and property type and indicate the exact location on an interactive map, which is driven by Leaflet and OpenStreetMap. All listing is available to the tenants who can filter by price and location, see the entire property details, save their favorites, and make a booking request. Booking lifecycle is under total control of the platform as the real-time notifications inform both sides about important actions. An owner and a tenant can then have access to a real-time chat feature which is Django Channels and WebSocket-driven and is available once a booking has been accepted. The two-way users are also able to post star ratings and

written comments about their rental experiences. The framework is developed with Django MVT architecture, MySQL as the database and Bootstrap as the frontend of the site and JavaScript as the interaction with clients.

1.2 Identification of client

The main client and beneficiary to Ghar Setu is a local property management agent, located in Itahari, Sunsari, Nepal, that is currently taking up rental enquiries, on behalf of several individual property owners. This client is a manual communication that allows landlords and tenants to interact through phone calls, face to face meetings at their office and paper rental contracts. The lack of an online platform implies that the information regarding the availability of a certain property is not digitally stored, a potential tenant must visit the office in person to inquire about the offers, and there is no organized method of tracing the state of a request to make a booking or to communicate or not.

The client stated that there was a definite need to have a centralized online solution where property owners will be able to list and manage their properties independently, tenants will be able to find and book the available homes without the need to visit the office and all communications and booking decisions to be digitally stored and tracked. The operational reality of this client is considered in the design of Ghar Setu, so the number of features, workflow, and the complexity of the interface are suitable to different users based on their digital literacy level.

1.3 Problem Scenario

The existing system of renting a house in the target market is disjointed, ineffective and very reliant on personal contacts and physical presence. The owners offering accommodation to rent out property put advertisement by way of hand-written signs on neighborhood boards, flyers printed and handed out in local areas, or word of mouth announcements via community contacts. This approach drastically restricts the visibility of both listing or in other words, a lot of existing properties do not find tenants who may make a perfect match.

Rental home seekers have an equally hard time. Lack of a central source of information makes them move to various places to view the properties, most of the times they arrive at the place only to find that the units that they heard about are occupied or do not correspond to what they had been told. It is not a secure method of assessing properties,

knowing the prices, or even evaluating the quality of a landlord prior to entering a contract. The owners and potential tenants do not have any kind of structured communication whatsoever, they only communicate by phone in an informal way with no documentation of what was said or agreed upon.

The very process of booking, when a tenant shows interest in a property, also presupposes additional face-to-face visits, verbal agreements, and handwritten contracts that can be easily disputed or misunderstood. It does not have a notification system that would inform the owners that someone is interested in their property, and there is no system to track the status of their requests by tenants. There is no form of reviews and rating on properties or landlord and therefore tenants cannot make informed decision based on experiences in the past. All these compounding issues prove that Ghar Setu is the necessary solution to have an organized, technology-intensive solution that would cover the entire lifecycle of the rentals process, starting with the discovery and ending with the constant post-booking communication.

2 Literature Review/Background Study

The chapter offers the theoretical background and contextual background that guided the design and development of Ghar Setu. It starts by defining the general project timeline, which identifies the major stages starting with the initial planning and requirement analysis all the way to design, development, and testing. The system in the chapter is then discussed as a viable provision to the recognized issue, how Ghar Setu responds to gaps and inefficiencies that exist in the current processes of home rentals.

Moreover, it has clear cut aims and objectives of the project so that the direction and anticipated results can be properly understood. Similar existing systems are also discussed in the chapter, providing critical comparison to find out their strengths and weaknesses, as well as reasons why the suggested solution is necessary. It also analyzes the possible risks, resources needed and technical background applicable in the project, giving a thorough insight into the environment the system is being built.

All in all, the research of the literature proves that Ghar Setu will be based on a highly investigated and informed knowledge of the existing practices, trends, and challenges in the home rental technology field, which will facilitate the creation of the reliable, efficient, and user-focused system.

2.1 Timeline of report/Project Gantt chart

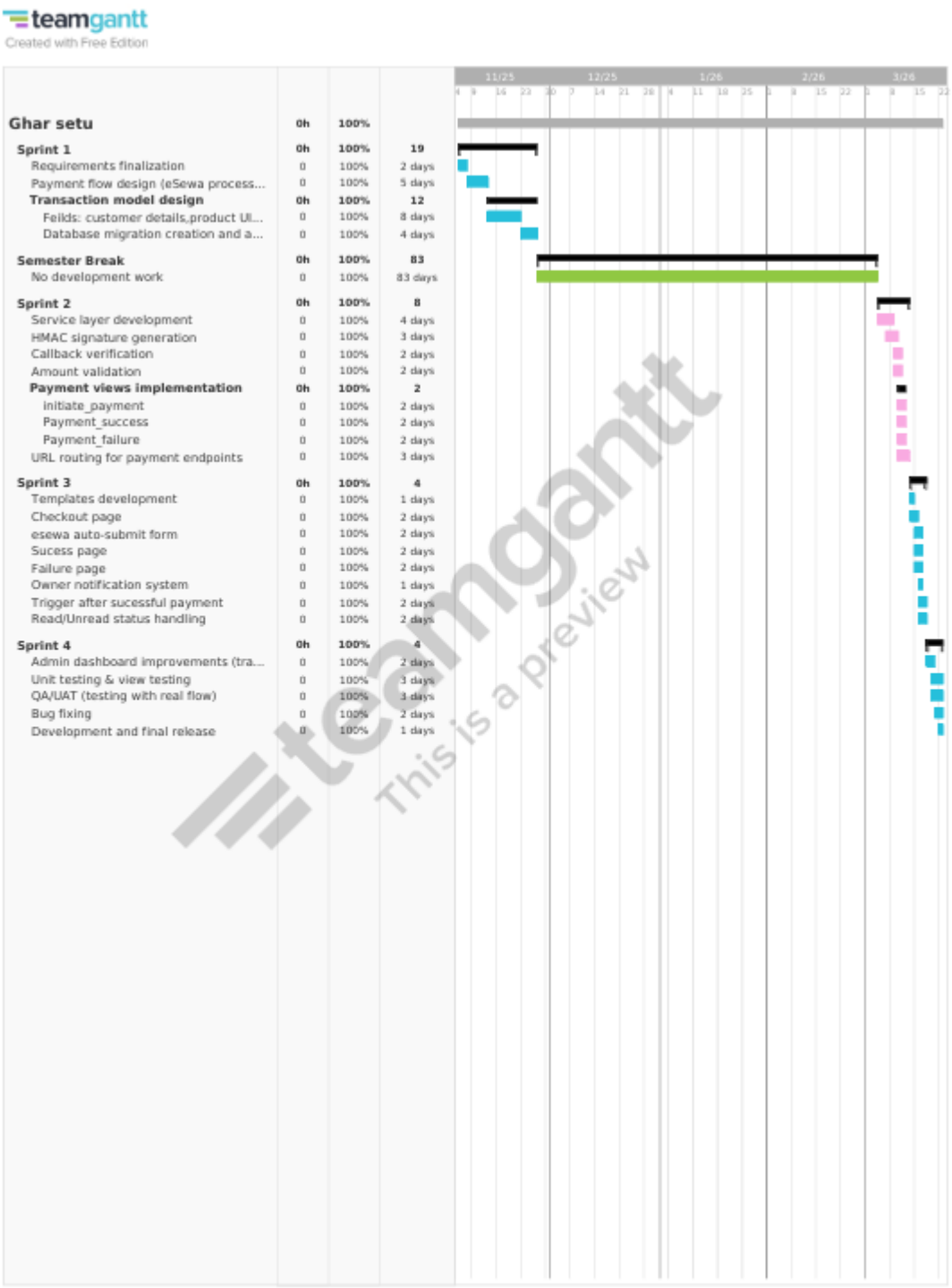

Figure 1: Gantt Chat

2.2 Project As a Solution

The fundamental inefficiencies that Ghar Setu addresses directly in the problem scenario revolve around the fact that the offerings create a single and unified digital platform that encompasses all rental processes. Instead of using physical announcements and the power of word-of-mouth promotion, the property owners can develop in-depth online posts that will be instantaneously visible to all the registered tenants visiting the site. Each entry also has several pictures, a complete description of the property, rental price, type of the property, contacts, and the exact location of the property on an interactive map based on Leaflet that allows tenants to have all the necessary information without having to visit the location in person.

Besides making accessibility, the system considerably increases the level of search. The filtering system will give the tenants the opportunity to filter their search results based on location and price range, and it saves a lot of time and effort to locate a good place to rent. This does not only enhance the experience of the users but also enhances the chance of favorable matches between tenants and the owners of the properties. The favorites feature gives tenants the option of saving the properties they are interested in, so that they can return and view them easily in future as one of the issues that affect this is that the tenants must remember or write down the various properties that they have previously viewed.

The booking request module brings in order and clarity into the process of negotiating a rental as it offers a formal and traceable way of showing interest in a property. The notifications help property owners to be notified about the new booking requests immediately and tenants to be notified about the acceptance or rejection of their requests in time. This minimizes doubt and time wastages that are prevalent in the traditional rental systems.

Besides, the post-acceptance chat feature removes the use of external communication tools and provides a secure, in-platform messaging service, which ensures the organization of all communication, as well as its easy accessibility. Lastly, the ratings and reviews system also present an essential element of trust and responsibility, as it enables the reviewers to leave feedback that is based on their experiences. This is a social proof mechanism that is completely lacking in the conventional manual process and gives tenants much power to make better decisions and motivates property owners to uphold high qualities.

Overall, Ghar Setu is not only digitizing the rental process but also increases the level of transparency, efficiency, and user trust, which makes it an all-encompassing and innovative solution to the problems found in the current one.

2.3 Aims and Objectives

The primary goal of the given project is to design, develop, and implement a fully functional web-based apartment rental management solution, which can streamline the overall process of renting property by both property owners and tenants, enhance the level of information exchange and transparency between the two parties, and offer a more accessible and, importantly, locally relevant alternative to the manual building rental system in use.

To reach this goal, the following specific objectives were determined:

- To carry out an in-depth study of the current rental procedure by interviewing the clients and conducting user surveys to determine the requirements of the system.
- To create a normalized MySQL relational database schema that would allow storing all user, property, booking, notification, message, and review data in a consistent and efficient way.
- To create a responsive web app that is built on Django MVT architecture, Bootstrap, and JavaScript and can be accessed by desktop and mobile browsers.
- To enforce secure user authentication, and role-based access control where property owner and tenant accounts will be distinguished.
- To create a complete property listing module with the possibility of uploading pictures, map-based location search with Leaflet and OpenStreetMap, and a description of the property.
- To introduce a booking request workflow enabling tenants to make requests and owners to accept or decline a request and send automatic notifications to both at every step.
- To create a real-time chat functionality using Django Channels and WebSocket that

will be made available between the owner and the tenant upon booking acceptance.

- To develop a favorites module where the tenants can create and maintain a personal list of preferred properties.
- To implement a ratings and reviews module enabling users to leave star ratings and written feedback on completed rental interactions.
- To conduct systematic functional and user acceptance testing of all modules and verify that the system meets the defined requirements.

2.4 Outcome and Deliverables

The desired results of the Ghar Setu project are a fully functional Django web application with a MySQL database in the background to maintain a high level of reliability, scalability as well as managing data efficiently. The system will be characterized by responsive and user-friendly interface which will be available on both desktop and mobile platform, giving smooth user experience to all users. It will have a full suite of fully functional modules, such as property listing and management, tenant search with advanced filters, booking request and management, live notifications, in-platform chat security, favorites system, and a comprehensive rating and reviews system.

Besides the functionality of the system core, the project is anticipated to reflect the appropriate front-end and back-end integration, good software development approach, and the ability to handle user data and communication. The application will also be put to test to be used, perform and reliable to typical usage scenarios.

The project deliverables are the full and properly documented source code that lies on GitHub, and Django model definitions and database migration files needed to restore the system. It also contains MySQL database schema and sample data scripts to facilitate testing and demonstration. The deliverables of the design include Figma wireframes and in-depth interface design documents that describe the appearance and user experience of the system.

Moreover, the documentation package contains entity relationship diagram (ERD), use case diagrams, and flowcharts that are visibly similar to the system design and behavior. There will be a detailed software requirement specification (SRS) document that will

detail the functional and the non-functional requirements of the system. The validation of system functionality will be demonstrated by the test case documentation, which will be accompanied by the related results.

Other deliverables involve user manual that could give proper guidance to property owners and tenants, and thus, easy navigation and comprehension of the features. Individual roles and responsibilities during the project will be recorded in group contribution records. Conclusively, this entire project report along with the screenshots of all the major system screens and features will give a comprehensive report of the design, development and implementation of Ghar Setu.

2.5 Similar Projects

There are three currently existing systems in the online property renting field reviewed during the research. These systems were helpful in offering good reference points on feature design, choice of technology, and definition of the gaps that Ghar Setu is meant to address.

Project 1: Nepal Property Bazaar

Hamro Property is a Nepali real estate listing portal through which real estate agents and property owners can list their properties either on sale or rent in the different cities within Nepal. The site contains property advertisements with photos, price, location and contact details of the listing agent. It further offers a simple search and filtering feature and featured listings section to premium advertisers *(nepalpropertybazaar, 2026)*. One of the most locally applicable comparators of Ghar Setu is Hamro Property since both companies work within the same geographical and cultural context. It is however limited to listing portal model. It does not have an inbuilt booking system, no tenant to owner messaging, no notification system, or ratings and review system. The site basically imitates an online classified advertisement board which does not facilitate any further steps in the rental process.

Project 2: Magic bricks

Magic bricks is one of the big Indian real estate's operating both rental and sales market. It provides certified property listings, an in-app request to contact system, tenant led management to agents, and a rudimentary messaging platform to introduce contact between parties. Magic bricks have strong property inventory volumes, and listing information that is supported is of good quality in terms of high-resolution photos, locality ratings, and information about amenities in the vicinity. Nevertheless, it is a discovery and first touch platform, but not a full-fledged rental management platform *(magicbricks, 2026)*. It lacks a proper booking request and acceptance process, lacks real time chat post booking, lacks a favorites module that has its own management interface and even its ratings system is limited to property level ratings with no option to receive continual feedback on the tenant's experience. It also targets a big commercially active market and has features like builder advertisements and investment analysis which is not applicable in the smaller and localized scenario of Ghar Setu.

Project 3: Airbnb

Airbnb is a well-known short-term rental service that offers complete end-to-end booking experience, featuring detailed property listings, a robust search and filtering system, a complete booking request and payment process, two-way review system and in-platform communication between hosts and guests. The competitive advantage of Airbnb is its refined user experience, trust and safety features and extensiveness of its booking management platform *(airbnb, 2026)*. However, Airbnb cannot be expected to suit long-term residential rental like long-term holiday and travel accommodation. It is not suitable for the monthly residential rental transactions that Ghar Setu is created because of its payment infrastructure, pricing model, host qualification requirements and service fee structure. Also, Airbnb does not have much local viability on Nepali property proprietors trying to connect with tenants in their community, as it is global and has complexities and charges that small individual landlords in Nepal cannot be expected to navigate.

2.5.1 Similar System Comparison

An eminent comparison of the three reviewed systems with HomeRental based on major criteria is provided in the following table:

Table 1: Useful comparison table format

Criteria	Nepal Property Bazaar	Magic bricks	Airbnb	Home Rental Management System
Target Market	Nepal	India	Global	Nepal (Local)
Listing	Yes	Yes	Yes	Yes
Map- Based Location	Basic	Yes	Yes	Yes
Free to Use	Yes	Yes	No	Yes
Real-Time Chat	No	Limited	Yes	Yes

2.5.2 Critical Evaluation

The critical analysis of the three systems reviewed has proven that all the systems are helpful in certain aspects, but all of them do not meet the needs of the system mix that has been found in Ghar Setu. The nearest to the local comparisons is Hamro Property but its feature range is too basic, as it is basically a digital noticeboard. Magic bricks is a more comprehensive feature that but its size and commercial focus would not suit the small, locally focused rental environment of the project. Airbnb is nearest to the end-to-end vision of rental management in terms of booking workflow and communication capabilities, but the short-term accommodation model, price structure infrastructure, and commerciality of this specific business model render it inappropriate as a residential rental tool in Nepal.

The unifying and missing feature of all three structures is the lack of an integrated end-to-end rental workflow that is both free and locally oriented as well as specific to long-term residential rental. The same can be said of Hamro Property and Magic bricks as they stop now of initial contact and the booking, communication, and review processes are left to the platform to handle completely. Airbnb handles these phases holistically yet in a situation which is not applicable in this project. Ghar Setu is perfectly poised to bridge this gap since it will have the local relevance of Hamro Property along with the completeness of workflow of Airbnb in a system tailored to the Nepali residential rental market.

2.6 Review Summary

The background analysis and review of the comparative system have established that there is a massive opportunity in the Nepali market of a locally oriented all-encompassing home rental management system. The review of the existing platforms shows that none of the propositions available is sufficient to accommodate the needs of the small property owners and tenants in a local Nepali environment that will need a free, integrated, and user-friendly platform that will take them through the entire lifecycle of the rental process. Technology decisions in Ghar Setu which are Django, MySQL, Django channels, Leaflet and Bootstrap are highly justifiable through technical strategies employed in successful similar platforms. The review has also found the significance of real-time communication, notification system and ratings and reviews as the various distinguishing features that make Ghar Setu more than a mere listing portal. The insights formed the basis of the feature scope, database design and workflow choices as explained in the following chapters.

2.7 Project Risks, Threats, and Contingency Plan

Any software development initiative, especially one with this kind of technical underpinning, has underlying risks which should be determined, evaluated and then mitigated before it occurs. Such risks can be caused by technical problems, the alteration of requirements, or shortage of resources, and unexpected external circumstances that can affect the overall success of the project. Discussing such uncertainties during the initial stages of the development process will aid in mitigating the possible disruptions and will allow conducting the project with fewer problems.

All significant risks related to the development of Ghar Setu are recorded in the following section and sorted out with each risk being elaborated with the explanation of the cause of the risk, how it is likely to affect the project, and what is the level of seriousness of the risk. The risks are properly analyzed to give a clear picture of the way each could negatively impact the system performance, development schedule, or user experience if it is not handled accordingly.

Moreover, a mitigation strategy is also considered relative to every risk that has been identified to mitigate on the likelihood or its impact. An elaborate contingency plan is

also contained to describe the measures that would be adopted in case certain risks are realized in development or deployment. The last section is the risk summary table, which gives a syncretic view of all the risks identified and their severity and mitigation measures, which can be easily referred to and risk monitored accordingly during project lifecycle.

2.7.1 Risk and threats

The following risks have been established because of review of the scope of the project, the technical complexity of Ghar Setu, the composition of the team and the environment within which the operation of the development period was to be conducted. Each of the risks is described separately with the cause, potential impact, and its severity.

Risk 1: Schedule Risk (Delays during Development)

Schedule Risk is at risk that the intended development tasks will not be done within the allotted sprints. Considering that Ghar Setu has technical scope, and it involves front-end design, Django backend development, MySQL database integration, Django Channels real-time chat setup, and Leaflet map integration, schedule overruns are among the most probable risks the project would experience. This is further added to by the fact that each of the team members is completing another academic course work.

- **Reasons:** Low estimation of the complexity of aspects like Django Channels WebSocket configuration and delivery of real-time notifications, sickness or personal emergencies of one or more team members, challenges in setting up the development environment of Django Channels including Daphne or Uvicorn server, unforeseen bugs that occurred during integration of the Leaflet map with the Django property model, and exam periods diminishing available time on development in the mid-sprint development cycles.
- **Potential Impact:** In case schedule risks are not managed, then the project will deliver on the submission deadline with some important features like the real-time chat module, notification system, or a booking workflow being partially implemented or not present during the final demonstration. The tests and documentation can be in haste, and the quality of the submission can be lower, and the grade will be directly influenced by this.

- **Seriousness:** HIGH - any week delay in Sprints 5 or 6 will ripple out into partial chat or notification functionalities, which form the core of the Ghar Setu value proposition.

Risk 2: Technical risk (Django Channels and WebSocket Integration).

Django Channels is a higher-level component than traditional Django views and it needs to have extra server infrastructure such as an ASGI server like Daphne, a channel layer backend like Redis or channel layer in-memory, and properly configured WebSocket routing and the traditional Django URL configuration. Any misconfiguration on any of these levels may lead to the failure of the real-time chat functionality altogether, and the errors can be hard to understand in case they are dealing with novice developers to asynchronous Django.

- **Reason:** No previous experience in a team using Django Channels and ASGI setup, version mismatches between Django, Channels, and Daphne, problems with setting up Redis as a channel layer backend in a shared and/or limited hosting environment, failure of WebSocket connections due to improper CORS or proxy settings on the deployment server, and race conditions in the chat consumer logic that resulted in out-of-order or non-delivery of messages.
- **Potential Effect:** In case of failure of the Django Channels integration, the real-time chat will not be possible to provide which will eliminate one of the most unique and technically amazing features of Ghar Setu. This would either involve returning to a polling-based messaging strategy, which is far less impressive, or showing the feature only in a local development environment. The result of these two undermines the quality of the end submission.
- **Level of Seriousness:** HIGH -Chat in real-time through Django Channels is a unique feature of Ghar Setu. The problem of technical failure in this module directly affects the quality of the entire system significantly.

Risk 3: Data Loss Risk

Data loss is defined as inadvertent losses or corruption of project assets such as Django source files, MySQL database scripts, HTML templates, CSS style sheets and project documentation. The fact that all the development will be done on the personal laptop of the team members and there will be no centralized server means that the failure of any of the devices would destroy much of the work done on the project.

- Cause: Hardware crash, e.g., a hard drive crash on one of the developer’s laptops,

accidental loss of files or corruption of Git repository, corruption of MySQL database due to improper shutdown during running migration, loss of migration history files which are needed to rebuild the database schema, power outages which lost unsaved work in the development environment.

- **Possible Impact:** It is possible that because of the loss of the source code, migration files, or database schema scripts, the team might need to recreate days or weeks of work by memory. In extreme cases, when migration history becomes corrupted, the Django database state might be lost forever unless a new setup is created, and all model changes must be re-enforced. This may cause flawed or prompt delivery deliverables by the time of submission.
- **Weight of the Criticality Level:** HIGH - Django migration files are quite sensitive as their loss may render the process of reconstituting the database extremely hard. Data loss is a severe and rather life-threatening risk without an appropriate backup plan.

Risk 4: Human Error Risk

Human errors include errors committed by team members in the process of development, testing or database administration that add in or add corruption or present system behavior in documentation. Since there are five team members and they are working on a common code base with Django MVT structure the potential of accidental errors is high.

- **Reason:** A developer ran an invalid Django migration like dropping a table in the development stage, a team member used untested code and pushed it straight to the main code, a view returning the wrong response, and Django settings that were never configured correctly, like wrong ALLOWED_HOSTS or DEBUG left open in the code.
- **Potential Effect:** Django has a tightly coupled MVT architecture, meaning that any human error may cause functional bugs in several modules at the same time. Broken forms, views, and templates are a result of errors in model definitions. Leaving the DEBUG mode on during production reveals much of an error information to third parties which is a security risk, as well as a quality risk. Any documentation mistakes may lie to the assessors about the capabilities of the system.
- **Severity:** MEDIUM -The human errors ought to be taken during any type of group development project but can be mitigated significantly with the introduction of enforced Git branching policies, code review prior to merge and a formulated testing checklist implemented at the completion of each sprint.

Risk 5: Hosting and Server Failure Risk.

The risk of server failure is associated with the fact that the environment that Ghar Setu will be deployed to demonstrate and evaluate its functionality can become unreadable, unconfigured, and unresponsive. Django applications have server setup needs, which do not match normal shared hosting setups, and Django Channels especially needs an ASGI-capable server.

- **Reason:** Outages or maintenance of cloud hosting providers like Railway, Render, or PythonAnywhere, incorrect configuration of the ASGI server Daphne during deployment of Django Channels, failure of the database connection in MySQL due to incorrect credentials or host configuration in the Django settings file, failure of the application to serve static files due to setting DEBUG to False in production necessitating the use of White Noise or an independent static file server, and resource constraints of free hosting plans making the application crash under the weight of very low load.
- **Potential Effect:** If the implemented system is online or misbehaves when the project is being demonstrated, the team might fail to demonstrate that the system is operational as planned. It is especially essential when it comes to real-time functionality, including chat and notifications, which need a live server setup with support of WebSocket. Failure to deploy within a short period before the assessment date does not leave a lot of time to recover.
- **Severity:** MEDIUM One should not confuse the deployment of Django Channels with the simple deployment of Django hosting; it is a more complex and challenging task to fully test the deployed environment early on to prevent last minute surprises.

Risk 6: CYBER Threat and Security Risk.

Ghar Setu has personal user information, property information, booking information, personal messages, and user reviews stored in a MySQL database available over a public facing web application. Such data profile predisposes the system to typical attacks on web applications and lack of security implementation on the development part might put users in danger.

- **Cause:** SQL injection MySQL SQL by inadequately validated form inputs SQL by using the Django ORM SQL by Cross-Site Scripting (XSS) SQL by Cross-Site Request Forgery (CSRF) SQL by unauthorized access to the Django admin interface SQL by

WebSocket connection hijacking trying to intercept or inject chat messages by using the Django Channels layer SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by attackers SQL by

- **Potential Impact:** An effective SQL injection or XSS attack will expose all the personal data of the registered users such as name, contact details, and personal messages. Hacking into the Django administration interface would provide an attacker with complete read and write privileges to the whole database. Interception of the chat messages would constitute a gross violation of the privacy of the users, and it may lead to disclosure of the contractual agreement between owners and tenants. All of these might lead to a legal liability and make the platform and use of the latter unsuitable.
- **Severity:** HIGH - Web security vulnerabilities need to be dealt with prior to development. Django includes inbuilt CSRF protection and SQL injection protection by ORM which are designed to be used properly and not compromised using an unprocessed SQL query or unsafe template syntax.

Risk 7: Team Co-ordination and Communication Risk.

As the five members in a team are working on various layers of a Django application at the same time, coordination and communication risk is an issue that has been practically implemented and is an ongoing problem throughout the project. The MVT architecture assigns work to models, views, templates, URL settings and static files such that poor coordination may result in the creation of overlapping changes, which will clash in the integration process.

- **Result:** Two developers concurrently adjusting the same Django model model and therefore creating incompatible migration files, a frontend developer writing a template with an expectation of a context variable that the view does not supply, inconsistent naming rules across URL patterns and template references, leading to 404 errors that are hard to debug, a backend developer making alterations to an API response format without notifying the frontend developer whose JavaScript relies on that format and tasks being left unfinished due to unable to determine who owns what in the sprint backlog.
- **Potential Effect:** The failure to coordinate in a Django project can lead to a system

where templates fail with missing context variables, URL reversals fail with renamed paths, and that history of migrations ends up not being consistent across local databases of the various team members. End of sprint integration sessions may be very time-consuming when the coordination has been lax at best during the sprint and therefore, they have little time to test and document.

- **Severity:** MEDIUM-- This risk is mitigated substantially by the fact that with a proper branching strategy, a common task board, and a weekly sprint review, the use of Git will still decrease this risk, but active communication regarding the interface contracts between frontend and backend developers needs to be upheld throughout the project.

2.7.2 Contingency Plan

To address the risks identified as well as reduce their effects on the delivery of the project, the following contingency plans have been drawn up and put in place within the entire process of developing Ghar Setu:

- Every development session (but not every development session) ends with all source code, such as Django models, views, templates, URL configurations, and migration files, being committed to a private GitHub repository, so that there are complete version history and the ability to roll back to any working state.
- The MySQL database schema and sample data are also dumped as SQL dump files weekly and sent to a different cloud storage service, which means that even if the development machine is lost or damaged, the database can still be recreated.
- Django Channels and the ASGI server setup were configured and tested on Sprint 1 on a dedicated test environment and then no other functionality was added on top of it, meaning that the most technically risky part was validated at the beginning of the development cycle.
- ASGI-supported on a cloud platform was deployed in Sprint 2 and would be used in further integration testing during the project, avoiding the chance of finding out about deployment-specific problems only at the end of submission.
- Two-week buffer was factored into the project schedule with a specific aim of absorbing any unforeseen technical issues, integration delays, or rework partially due to client feedback or requirement changes.
- Django has inbuilt CSRF security, all forms are being secured with CSRF protection,

ORM is utilized to perform any query on database to avoid SQL injection vulnerability, and all user input in templates has escaped to avoid XSS vulnerability through Django auto-escaping.

- There is a Git branching approach where individual developers are assigned a named feature branch and no code may be accepted by a merger to the main branch without at least one peer review to avoid untested or conflicting code making it into the common codebase.
- The weekly sprint review meetings, which are conducted weekly on a Friday, involve all the members of the team reporting what they have accomplished during the time slot, blockers, and agreements of the interface contracts between the frontend templates and the backend views of the next week.

Table 2: Simple risk management Table

Risk	Cause	Impact	Severity
Schedule Risk	Feature complexity, exam period, debugging delays	Incomplete features, rushed submission	High
Django Channels Integration	ASGI config, Redis setup, WebSocket routing errors	Chat feature fails, key differentiator lost	High
Data Loss	Hardware failure, migration corruption, accidental deletion	Work reconstruction required, incomplete deliverable	High
Human Error	Model mistakes, bad migrations, wrong Git merges	Cascading bugs across MVT layers	Medium
Server/Hosting Failure	ASGI config, static files, database connection errors	System inaccessible during demonstration	Medium
Cyber Threats	XSS, SQL injection, CSRF, admin exposure	Data breach, privacy violation, legal liability	High
Team Coordination	Conflicting migrations, unclear ownership, naming errors	Integration bugs, timeline overrun, template crashes	Medium

2.8 Resource Requirements

The hardware and software resources given were identified as important for success.

2.8.1 Hardware Requirements

Table 3: Hardware requirements

Hardware	Specification	Purpose
Development Laptop	Intel Core i5 or better, 8 GB RAM, 256 GB SSD	Django development, database management, testing
Smartphone (Android/iOS)	Any mid-range device with a modern browser	Mobile responsiveness and UI testing
Internet Connection	Minimum 10 Mbps broadband	GitHub, package installation, cloud deployment, WebSocket testing
External Hard Drive	500 GB minimum	Offline backup for source code and database exports

2.8.2 Software Requirements

Table 4: Software requirements

Software / Tool	Version	Purpose
Ubuntu 22.04 / Windows 10	Latest	Operating system for development
Python	3.11+	Core language for Django backend
Django	4.2 LTS	Web framework using MVT architecture
Django Channels	4.x	WebSocket and ASGI support for real-time chat
Daphne	4.x	ASGI server for serving Django Channels
MySQL Server	8.0	Relational database for all application data
MySQL client (Python driver)	Latest	Django database connection to MySQL
Gmail SMTP (Django email backend)	Latest	SMTP email delivery for booking notifications triggered by Django signals
Redis	7.x	Channel layer backend for Django Channels
Leaflet.js	1.9+	Interactive map for property location selection
Bootstrap	5.3	Responsive frontend CSS framework
Visual Studio Code	Latest	Primary code editor
Git + GitHub	Latest	Version control and team collaboration
Figma	Web-based	Wireframe and UI prototype design
Postman	Latest	API endpoint testing
Google Chrome / Firefox	Latest	Browser testing and developer tools

3 Methodology

This chapter explains the software development methodology chosen in case of Ghar Setu and gives a clear rationale of the choice that has been made with the help of scope of the technical aspect of the project, structure of the team, schedule and nature of the relationship with the client. An adequate and regularly implemented methodology will help to make the development process structured, quality-oriented, and flexible.

3.1 Considered Methodologies

Before making the final decision on the software development methodologies to be used in Ghar Setu, three methodologies were considered:

- **Waterfall Model:** This is a linear and sequential approach where one stage should be completed then the next one will start. Phases are not iterative since they follow a progression of requirements analysis, system design, implementation, testing, and deployment. Waterfall fits perfectly in project where the requirements are fully stable and known at the beginning of the project and changes are unlikely to occur after the development has begun. In the case of Ghar Setu, however, where some of the technical specifications, such as the specific configuration of Django Channels, and the integration of Leaflet map, could not be settled upon until post-exploratory development, the strict sequential nature of Waterfall would have been too inflexible to allow the learning and adaptation needed.

- **Agile Scrum:** A short, incremental approach to work organization, breaking it into short sprints of two to three weeks, each delivery of which includes an increment of the system that is both working and demonstrable. Scrum focuses on teamwork, frequent feedback to the client and adapting to changing requirements between sprints. Product backlog is used to manage all features and tasks, and daily standups are used to keep the team on track. In the case of Ghar Setu, where the client has local needs and the specific characteristics of the product are technologically complicated and require a continuous improvement in their definition, and the enterprise includes five employees with different roles, Agile Scrum offers the perfect combination of organization and flexibility.

- **Prototyping Model:** This is a method whereby a prototype is created as rapidly as possible and subsequently enhanced on a certain basis through feedback from the user and client. Prototyping is effective especially when the user interface and user experience are the key design issues. Although it would be useful in investigating the Ghar Setu interface design during early sprints, prototyping would not give enough structure to manage the complexity of the backend development, database, real-time channel architecture, and testing that the Ghar Setu would entail.

3.2 Chosen Methodology: Agile Methodology

The development methodology that was adopted to develop Ghar Setu after considering all the three methodologies will be Agile Scrum. Five main considerations were used to make this decision. First, the project has several technically different modules such as Django backend, real-time Django Channels chat, Leaflet map integration, MySQL database and Bootstrap frontend, which can be developed, tested and implemented across separate sprints instead of simultaneously. Second, the client can be contacted periodically during review meetings at the conclusion of each sprint, which is inherent to the Scrum sprint review ceremony and gives a possibility to verify and optimize requirements in the course of the work. Third, Agile enables the team to work on the most valuable features during the initial sprints to ensure that by the conclusion of the first several sprints one has a working system that is demonstrable and partially complete. Fourth, Agile is iterative, which provides the opportunity to develop high-risk technical areas like Django Channels earlier and avoid finding the fatal integration issues just before the deadline. Fifth, the formal sprint backlog and the retrospectives meetings offer the coordination model necessary to coordinate five team members with individual tasks and work on the same codebase.

The project was divided into seven two-week sprints. The project had a product backlog that was used to monitor all pending features and identify bugs. The planning of the tasks to be performed during the sprint was provided in the form of sprint planning at the beginning of every sprint and distributed the tasks to the individual team members. An update of brief standup was provided in the group on all working days to inform about the progress and raise blockers. End-of-sprint sprint review and retrospective sessions evaluated the work carried out during a sprint, what had to be improved and what planning was to be undertaken in the following sprint.

4 System Design

In this chapter, the whole structure and logic design of Ghar Setu is given. It outlines the structure of the system, the database design, how various user management roles operate with the system and the sequence of the most significant processes. These design artefacts acted as technical blueprints to guide implementation and assist in ensuring that the final system is coherent, maintainable and in line with the requirements that are specified.

4.1 System Architecture

Ghar Setu is written in the Django Model-View-Template (MVT) architecture that is the Django implementation of the traditional Model-View-Controller (MVC) pattern. The Model layer in this architecture will oversee defining the data structures, as well as all communication with the MySQL database via the ORM in Django. The View layer includes the business logic of the application itself, which receives the HTTP messages, data processing, and what response to send. The Template layer will render the HTML pages returned to the browser of the user, with the help of Django template language, to dynamically incorporate the information in the View into the HTML pages.

Ghar Setu at infrastructure level is implemented in a two-server setup. All non-real-time page requests are served by Daphne as an ASGI server and the Django Channels WebSocket connections are served by Django Channels real-time chat feature. MySQL is a persistent database that holds all the data of the applications. Redis is the channel layer backend to Django Channels and takes care of the routing of messages between WebSocket consumers. CSS style sheets, JavaScript files and uploaded images of property are served independently on White Noise middleware. The Leaflet.js script is loaded on the frontend and interacts with OpenStreetMap tile server outside the system to display the interactive map of the property location.

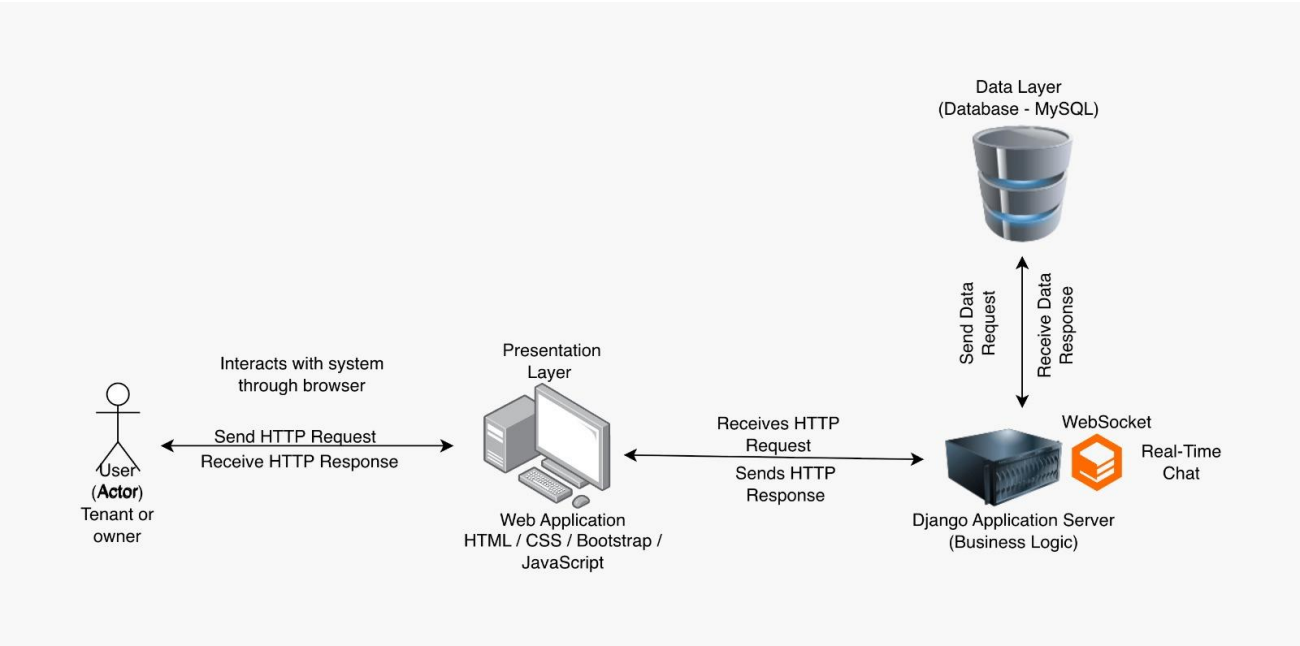

Figure 2: System Architecture

4.1.1 Email Notification Architecture (Django Signals and SMTP)

In Ghar Setu, email notification system is automated with the in-built email framework of Django with Gmail as the SMTP server. It is configured to transmit emails via the SMTP server of Gmail (smtp.gmail.com on port 587 and encrypted with TLS) and the sender credentials are saved in the Django settings file as environment variables instead of being hard coded in. Such a strategy would guarantee that the Gmail credentials are not revealed in the GitHub repository of the project.

Email delivery is automatically activated by Django Signals which is an inbuilt Django mechanism that enables decoupled components of the application to be informed of when certain activities happen in other parts of the system. Instead of implementing email logic as a part of views, the email notification logic is implemented as a part of signal receiver functions in a special signals.py file in the corresponding Django app. This design maintains a clean view code and makes sure that an email is dispatched in any case and consistently no matter what section of the codebase is invoked to perform the booking action.

There are three email notification events that are used in Ghar Setu, which are associated with the phases of the booking lifecycle. First, when a tenant makes a new booking request, a postsave signal is sent on the Booking model. The function of the connected receiver is to verify that the booking instance is a new instance (via the created flag that is offered by postsave) and sends an email to a registered email

address of the property owner telling the owner that a new booking request has been made against their listed property. The email will consist of the name of the tenant, the property name, the date when they want to move in, and a reminder to log in and see the request. Second, when an owner receives an update of a booking to Accepted the post save signal is again received on the Booking model. The receiver functional identifies that the status field has been updated to Accepted and dispatches booking confirmation email to the registered email address of the tenant. This email will testify that the reservation is confirmed and gives the contact information of the owner as well as the notices the tenant that the in-platform chat is operational. Third, in case an owner changes the status of a booking to Rejected, the same signal mechanism sends an email with a rejection notification to the tenant, letting them know that the request was not accepted and encouraging them to go search for other properties available on the platform.

4.2 Entity Relationship Diagram (ERD)

The following are the main entities and relationships in the ERD of Ghar Setu. The User entity holds all registered users with the following fields: username, email, password hash, phone number, profile image, and a field a role differentiating the owner and tenant account. The Property table contains all the rental listing information such as title, description, type of property, rental price, address, long and latitudes of the Leaflet map pin, availability, and a foreign key to the owner User. The Property Image entity holds the paths of individual image files associated with a property and thus a property could contain many images. The Booking entity stores tenant booking requests and has attributes of tenant requesting a booking, the target property, the requested move in date, the current state (pending, accepted or rejected) and the creation and the last update time. The Notification entity contains single notification messages of users with the fields being the recipient, the message content, a read or unread status flag and the creation date. The Message entity is used to store individual chat messages, where the sender, receiver, the respective booking, message contents, and the timestamp of when the message was sent are kept, which are used to support the post-acceptance chat feature. The Favourite entity establishes the many-to-many association among the tenants and properties they have saved. Review entity contains the reviews of users including the reviewer, reviewed property, star rating between one and five, review text, and time of a review.

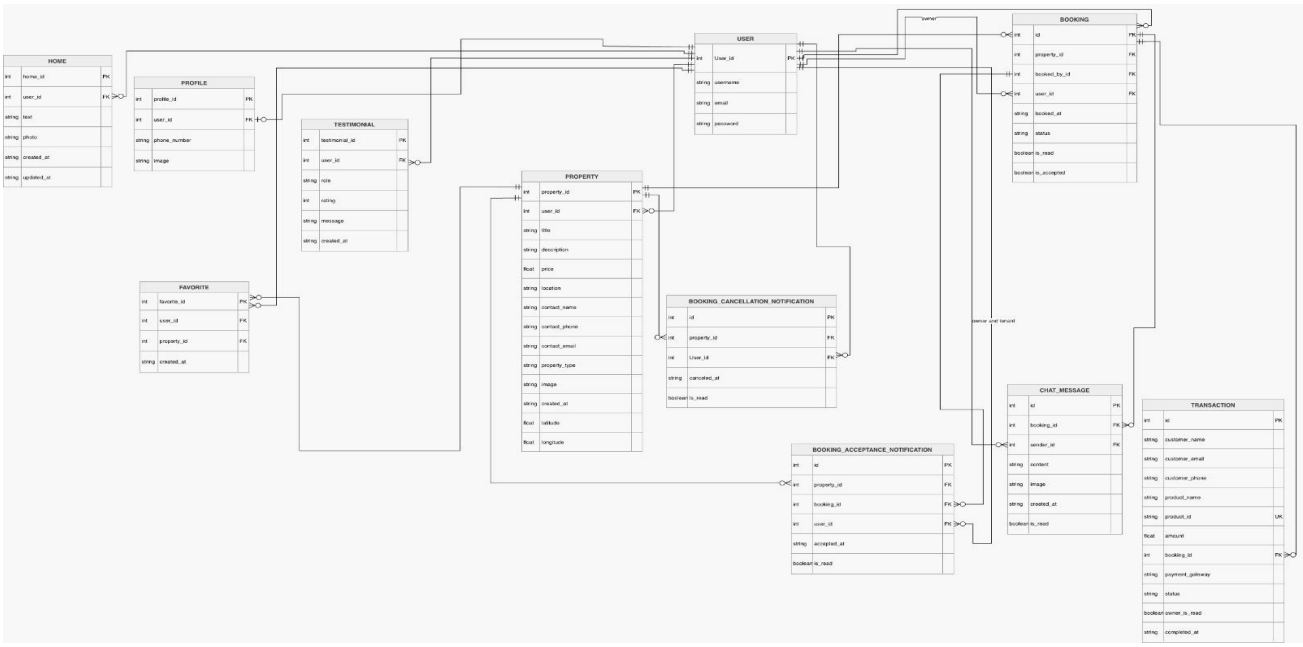

Figure 3: ERD

4.3 Use Case Diagram

The Ghar Setu uses case diagram that recognizes three actors, namely the Administrator, the Owner and the Tenant. The Administrator will have access to the Django Web-based administrative interface to handle the account of all the users, property listings, bookings and reviews and the notification of all the users on the platform. The Owner can create an account and log-in, create and manage property listings with pictures and map position, view and manage booking request received, accept or reject booking request, be notified on what the tenant is doing, chat with booking request accepted tenants and see his owner profile page. The Tenant will be able to register and log in, explore all the properties at their disposal, use search filters to filter down the results by location and price, view the detailed page about the property, save the properties in their favourites, make a booking request to the property owners, be notified about the booking choices made, chat with the owners after the booking acceptance, leave ratings and reviews on the properties, see the About page and Contact page, and manage their profile as a tenant.

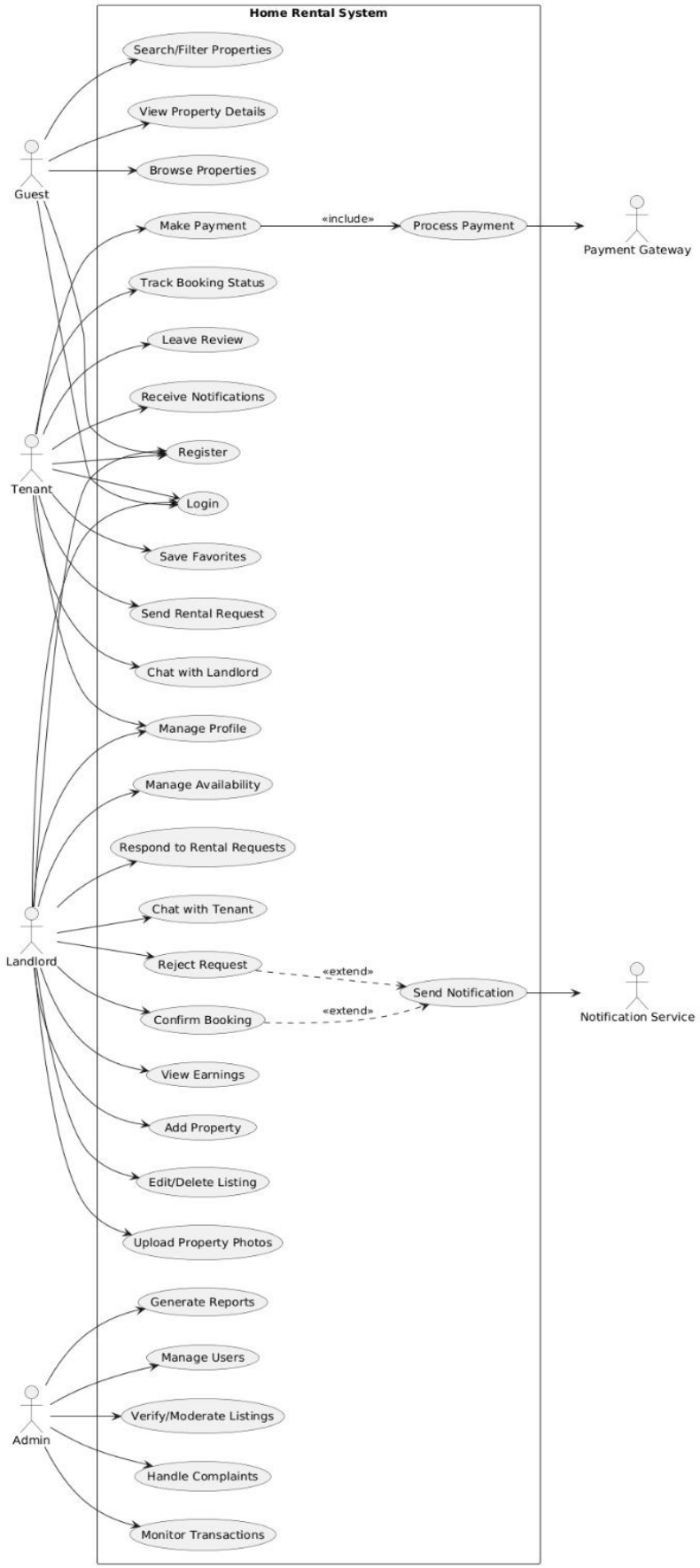

Figure 4: Use Case Diagram

4.4 Flowchart

The workflow flowchart of the booking is that the tenant accesses the Ghar Setu platform by logging in. The system will authenticate the session and redirect to the login page in case of invalid or expired session. The tenant then browses the property listing page after authentication and can apply filters to the property listing by location and price range. Tenant chooses an item of interest and sees the complete detail page with all pictures, Leaflet map pin of location, rental price, property category and contact details of the property owner. The tenant presses Book Now button and a booking request confirmation modal shows up. The tenant agrees to the request, and this is stored in MySQL in the Booking table with a status of Pending. A notification record is generated to the owner of the property, and the notification is sent in real time via Django Channels. The notification is received by the owner who goes to his bookings dashboard and looks at the request submitted by the tenant. The owner either clicks Accept or Reject. In the event of the booking being rejected, it is changed to Rejected status, a notification is posted to the tenant, and the tenant is told that the request was not successful. In case of the booking acceptance, the status is changed to Accepted, a notification is dispatched to the tenant, and an automatic activation of the chat interface between the owner and tenant is performed. The chat module that is powered by Django Channels allows both parties to exchange real-time messages. The tenant is also free to revisit the property page and leave a star rating and a written review at any point.

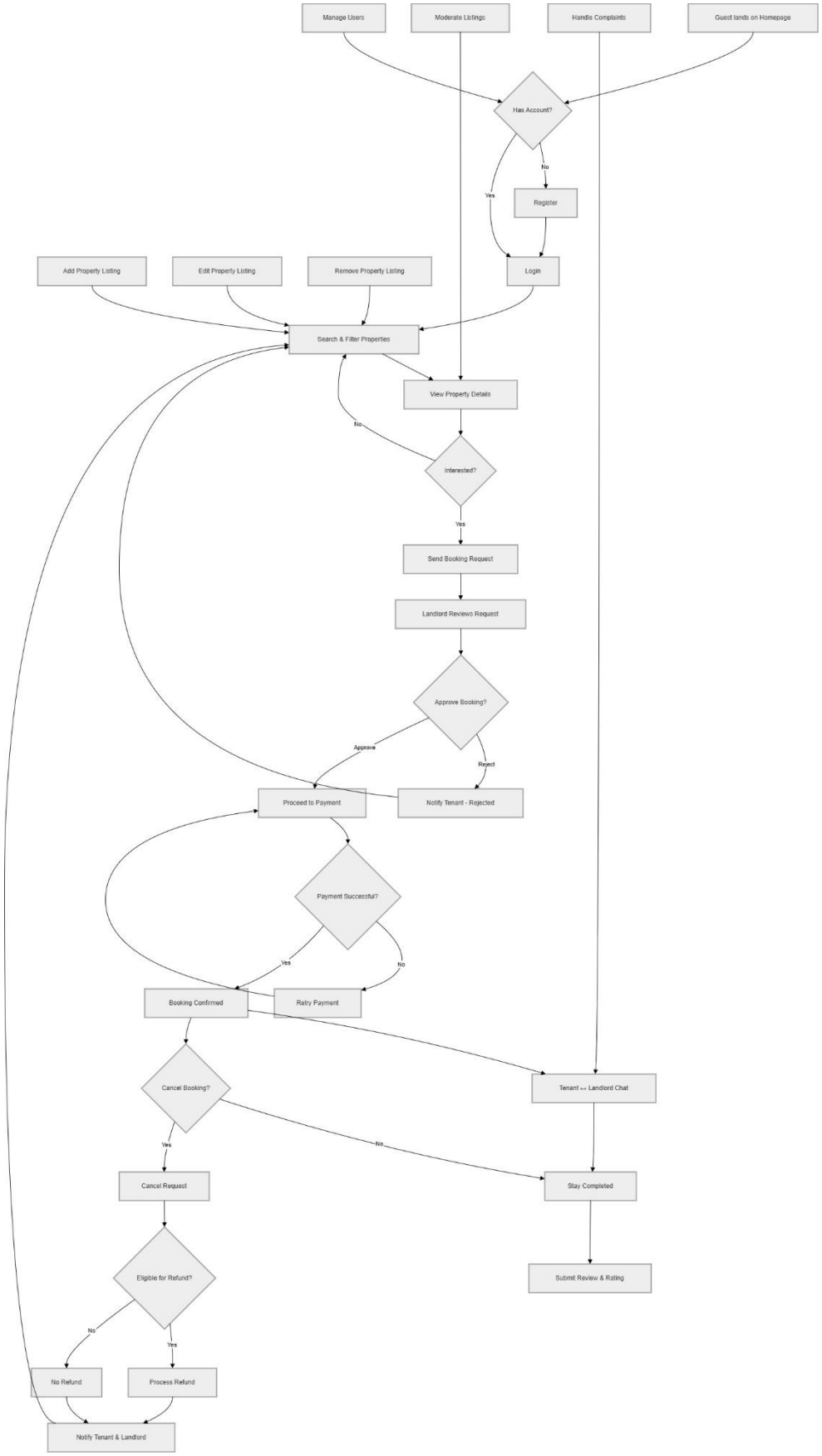

Figure 5: Flowchart

5 Software Requirements Specification (SRS)

5.1 Introduction

The main objective of this document is to outline the functional and non-functional requirements of the system Ghar Setu system. Ghar Setu is a web-based home rental system that bridged house owners and the potential tenants which allows seamless property listing, surfing, booking and can communicate.

5.2 Scope Ghar Setu is developed to:

- Let property owners post rental properties.
- Allow tenants to search and filter and book properties.
- Communication after acceptance of booking by chat.
- Favorites, support notifications, and reviews.

The system will streamline the simplification of the process of home rental through online website.

5.3 Definitions, Acronyms, and Abbreviations.

- **SRS**: Software Requirements Specification.
- **MVT**: Model-View-Template DB: Database
- **UI**: User Interface

5.4 Overall Description

5.4.1 Product Perspective

Ghar Setu is a single web app developed based on the Django framework based on the MVT architecture. It integrates:

- Backend: Django
- Database: MySQL
- Frontend: Bootstrap, JavaScript, CSS, HTML.
- Real-time communication: Django Channels.
- Integration of maps: Leaflet and OpenStreetMap.

5.4.2 Product Functions

The features of the system include the following:

- Registration and authentication of the user.

- The management of property listing.
- Filtering and browsing property.
- Booking management, Notifications system, Real-time chat, Reviews and ratings, favorites management.

5.5 Characteristics and Classes of users.

Table 5: Characteristics and Classes of users

User Type	Description
Admin	Manages system, users, and content.
Property Owner	Prepares and maintains rental property listings.
Tenant	Searching, Book, and Reviews Properties.

5.6 Operating Environment

- Web Browser (Firefox, Safari, Chrome)
- Operating Systems: windows, MacOS, Linux.
- Server: Django friendly hosting server.

5.7 Constraints

- It needs to be connected to the internet.
- Relying on third-party Map services (OpenStreetMap).
- Chat in real-time relies on WebSocket.

5.8 Assumptions and Dependencies.

The users possess rudimentary internet and browser knowledge.

Django and MySQL are both configured on the server.

The users present the correct property and personal details.

5.9 Functional Requirements

5.9.1 User Management

- Users are allowed to sign up using email and password.
- Users can log in and log out Users are able to edit profile data.
- Role based access (Owner/Tenant)

5.9.2 Property Management (Owner)

- Add new property listings Make amendments to current property.
- Delete property listings Upload property images Mark price, type and location through map.

5.9.3 Property Browsing (Tenant)

- See everything that is available.
- Filter properties by:
  - Location
  - Price and detailed property information.

5.9.4 Booking System

- Tenants can place booking requests.
- The owners can approve or decline requests.
- Status of booking is changed appropriately.

5.9.5 Notification System

- Send owner notification on receiving booking request.
- Inform tenants of acceptance/rejection of booking.

5.9.6 Chat System

- Only after the acceptance of the booking, the chat is turned on.
- Live chat between owner and tenant.

5.9.7 Favorites

- Tenants can save properties
- View saved properties later

5.9.8 Reviews and Ratings

- Tenants can rate properties Reviews can be left after the bookings.

5.9.9 Static Pages

- About page
- Contact/Help page

5.10 Non-Functional Requirements

5.10.1 Performance

- Pages should be loaded in 2-3 seconds.
- Assist more than one user at a time.

5.10.2 Security

- Authentication system that is secure.
- Password encryption Privacy against unauthorized access.

5.10.3 Usability

- User-friendly interface Adaptive design to various devices.

5.10.4 Reliability

- System should be available 24/7 with Minimal downtime

5.10.5 Scalability

- System must be able to accommodate more users and properties.

5.10.6 Maintainability

- Code must adhere to modular structure.
- Easy to update and debug

6 Testing

The Ghar Setu testing was done continuously during the development process with specific testing activities factored into the plan per sprint. Functional testing was to ensure that every module performs as per its requirement. The interaction between the Django views, MySQL database, Django Channels WebSocket layer, and Leaflet map components were confirmed by integration testing. The test was carried out using user acceptance by representatives of the client organization and a limited number of representative users of the locality. The results of the tests conducted in major system modules are reported in the following test cases.

6.1 User Registration

6.1.1 Creating Account

Table 6: Test Case 1: To create new account

Objective	To create a new account.
Input Data	All data were filled.
Expected Output	Account should be created successfully.
Actual Result	The account was created successfully.

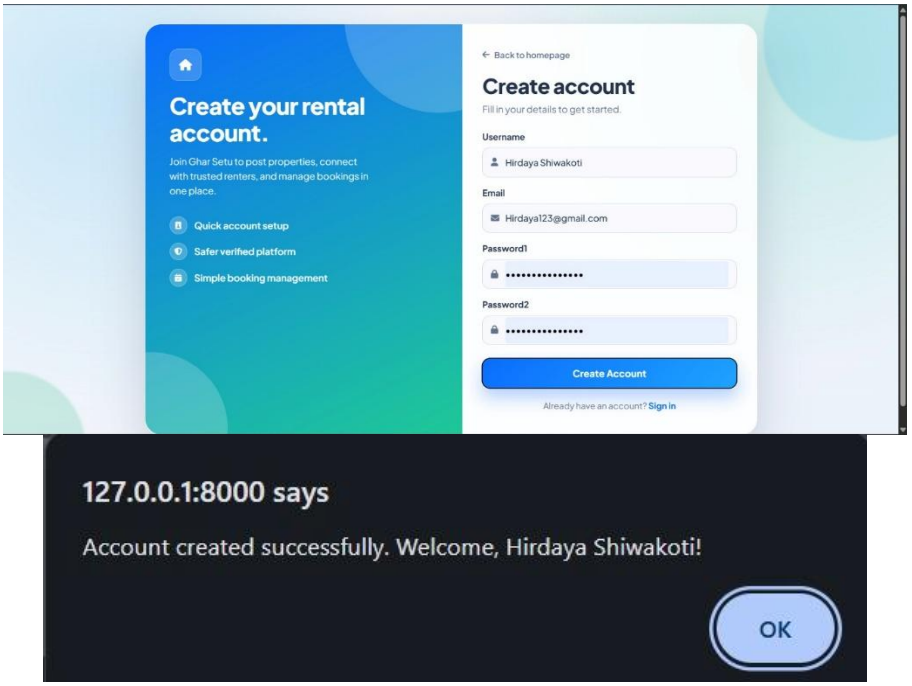

Figure 6: User Registered Successfully

6.1.2 User Login (with wrong credentials)

Table 7: Test Case 2 : Login with wrong credentials

Objective	Login with the wrong credentials.
Input Data	Wrong credentials were entered.
Expected Output	Account should not login.
Actual Result	Account was not logged in.

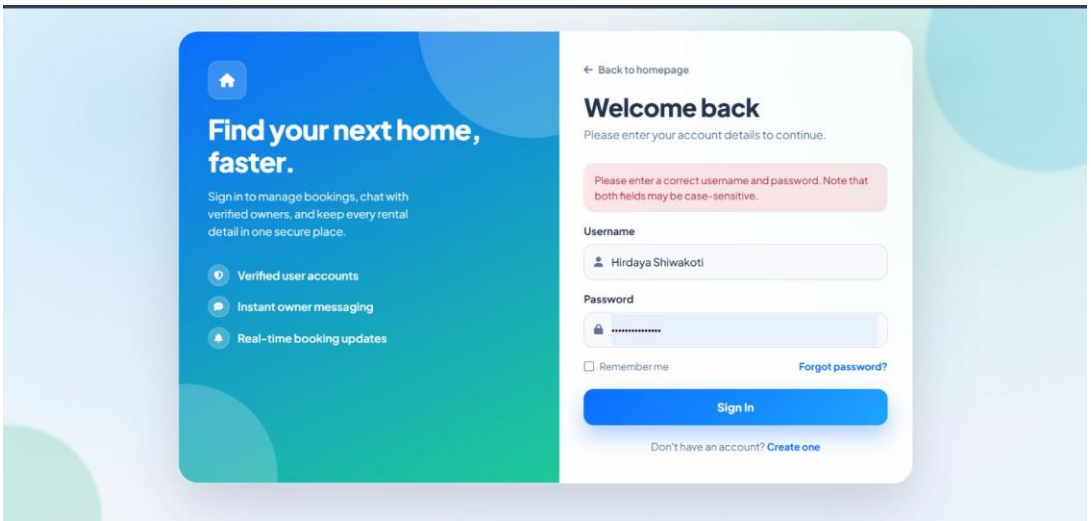

Figure 7: Login Failed

6.1.3 User Login (with correct credentials)

Table 8: Test Case 3: Login with current credentials

Objective	Login with the correct credentials.
Input Data	Correct credentials were entered.
Expected Output	Account should successfully log in and should show home page.
Actual Result	Account was logged in and home page was shown.

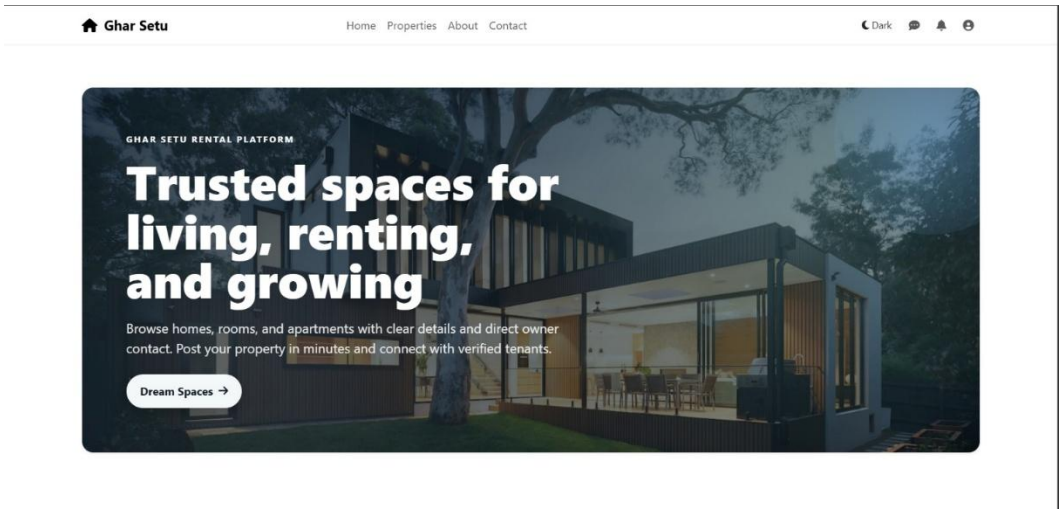

Figure 8: Login Successful and Home page opened

6.2 Property Listing

Table 9: Test Case 4: Property Listing

Objective	To list the property.
Input Data	All fields were fulfilled.
Expected Output	Property should be listed.
Actual Result	Property was listed successfully.

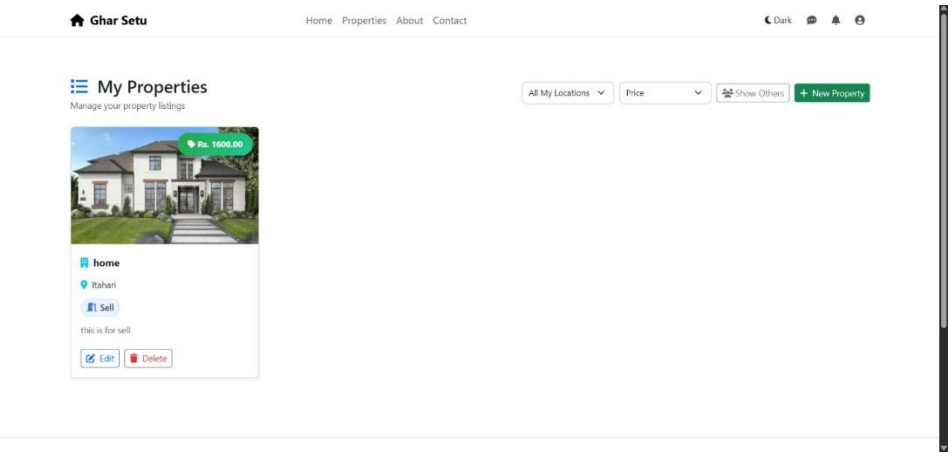

Figure 9: Property Listing Successful

6.3 Property Update

Table 10: Test Case 5: Property Update

Objective	To edit and update the fields.
Input Data	Added parts to be edited or updated.
Expected Output	Fields should be updated.
Actual Result	Fields were updated successfully.

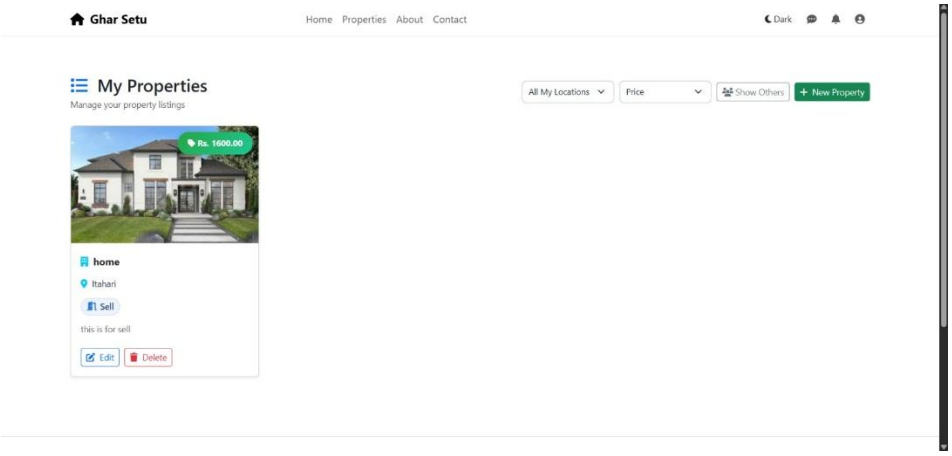

Figure 10: Interface before Renting

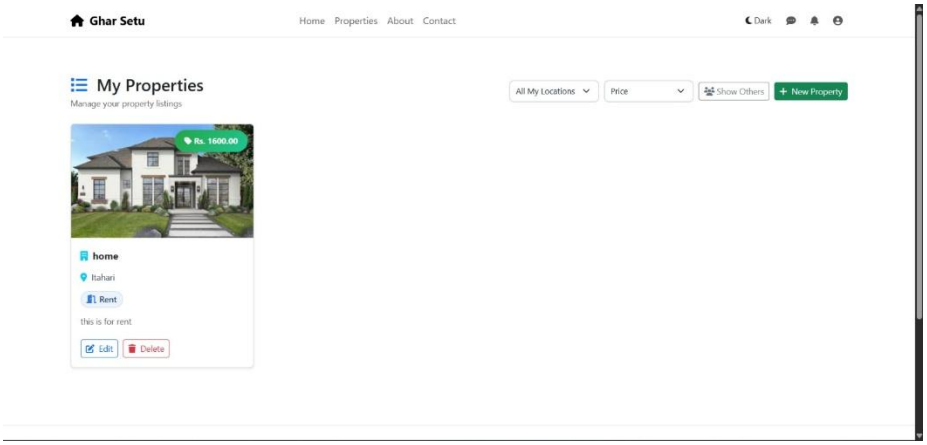

Figure 11: Interface After Renting

6.4 Property deletion

Table 11: Test Case 6: Property Deletion

Objective	To delete the property.
Input Data	Delete button was clicked.
Expected Output	Delete the existing property.
Actual Result	Successfully deleted the existing property.

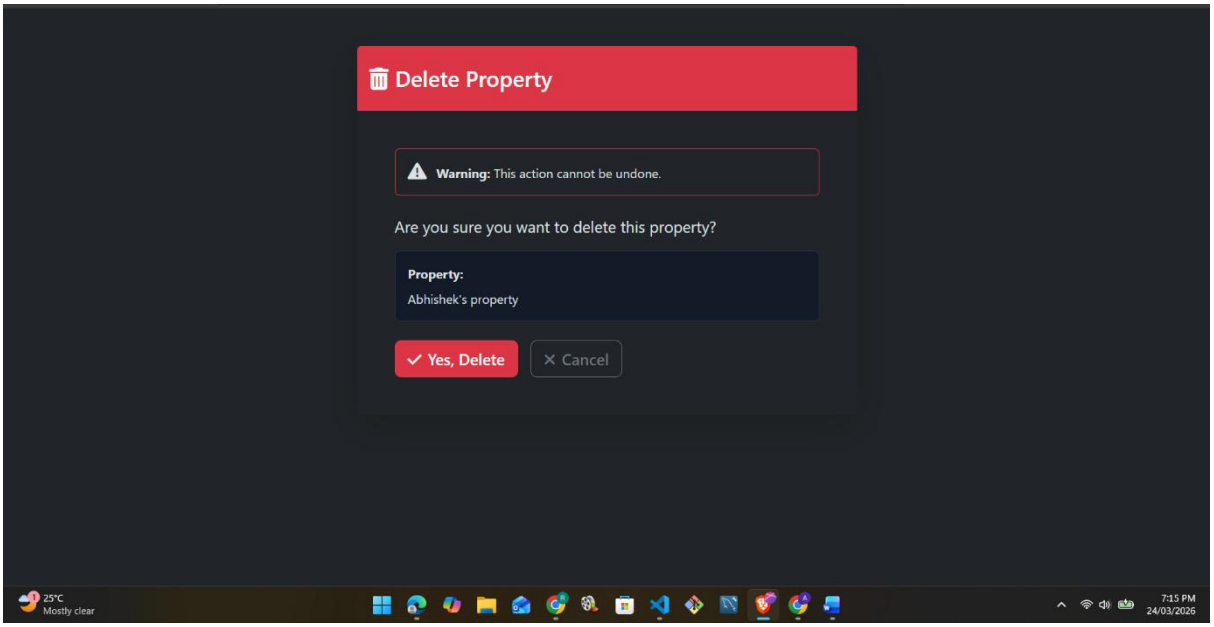

Figure 12: Property Deletion

6.5 Property Filter

Table 12: Test Case 12: To filter property

Objective	To filter the property according to the city.
Input Data	Filter button was pressed.
Expected Output	Filtered the city.
Actual Result	Successfully the filtration was done.

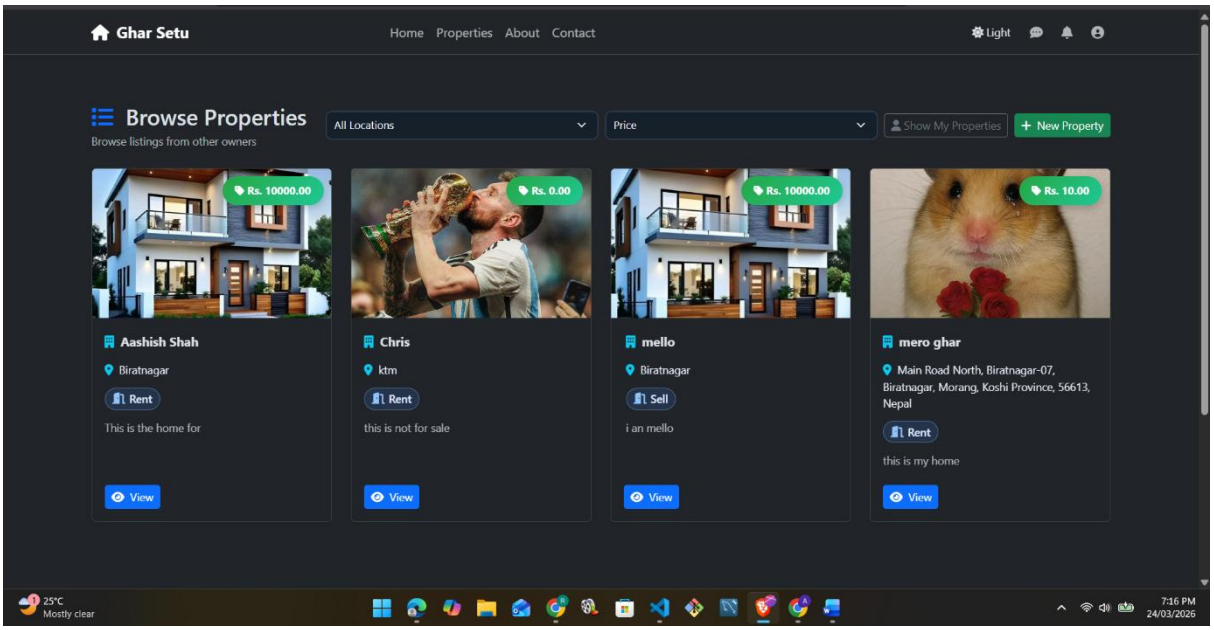

Figure 13: Property Filter

6.6 Booking request and Approval

Table 13: Test Case 13: Booking Successfully

Objective	To place a Booking and get it approved.
Input Data	Booking button was pressed.
Expected Output	Booking should be done.
Actual Result	Successfully booked.

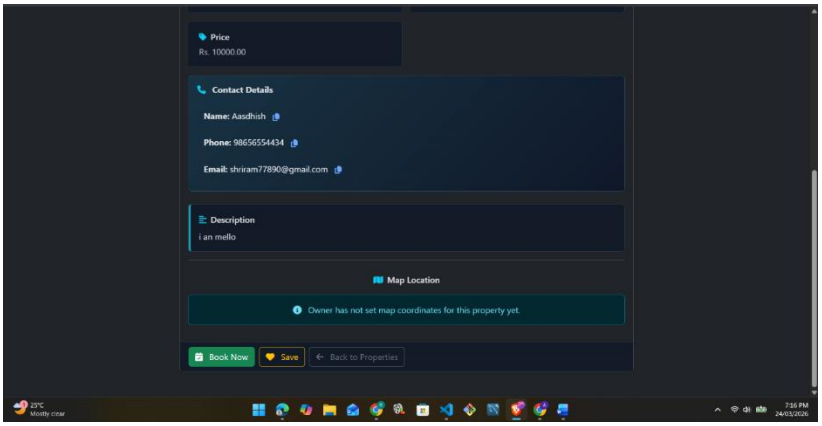

Figure 14: Booking Request

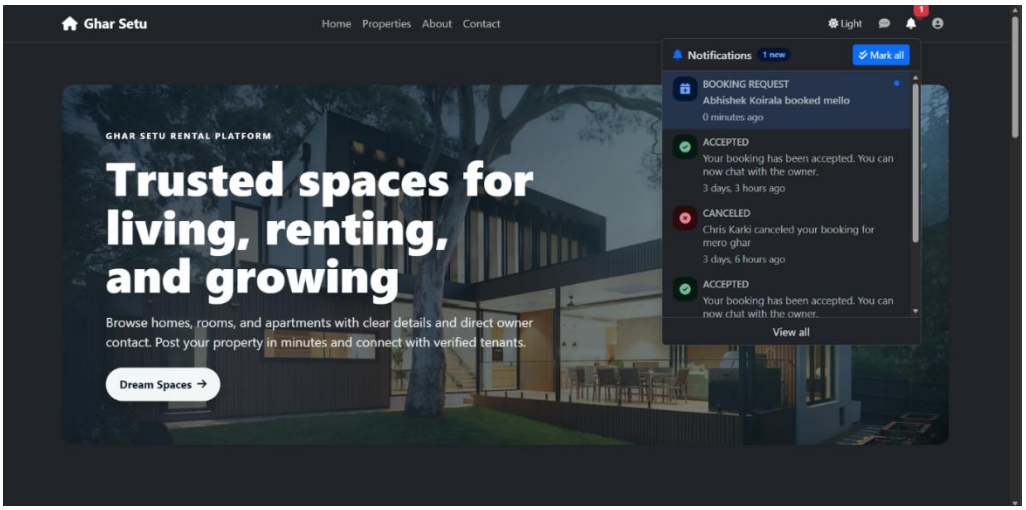

Figure 15: Notification of property

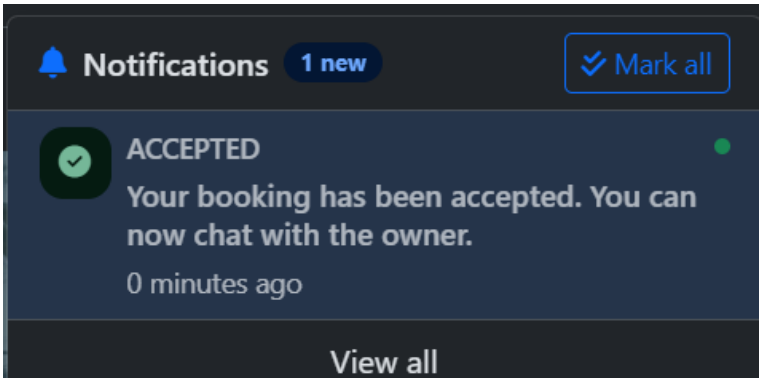

Figure 16: Booking Approved

6.7 Booking Rejection

Table 14: Test Case 14: Booking Rejection

Objective	To check whether the booking is cancelled
Input Data	Notification icon was clicked
Expected Output	Booking cancelled
Actual Result	Your booking was cancelled by the owner

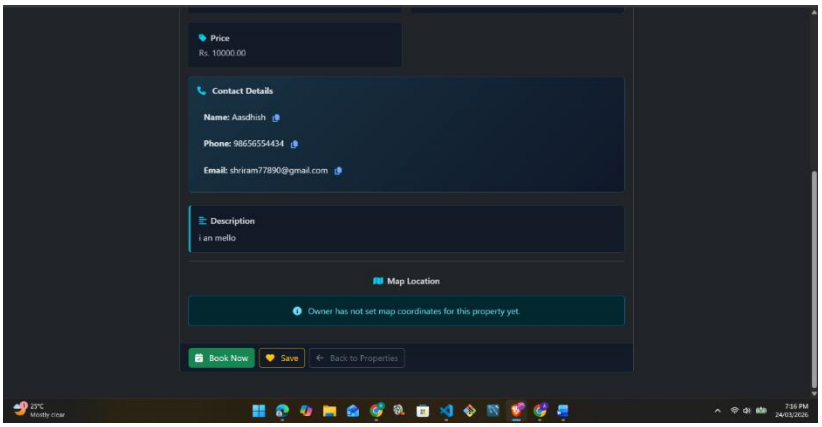

Figure 17: Booking Request

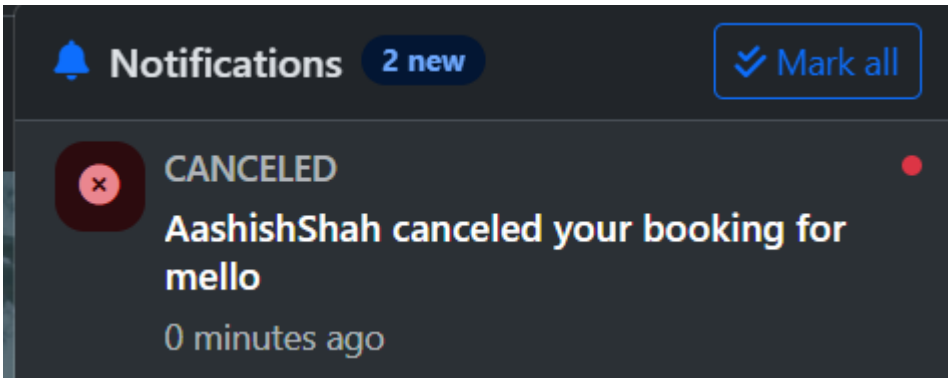

Figure 18: Booking Cancel

6.8 Payment Failure

Table 15: Test Case 15: Payment Failure

Objective	To place the payment and check whether the payment was failed.
Input Data	All the payment was done.
Expected Output	Payment should fail.
Actual Result	The payment was failed.

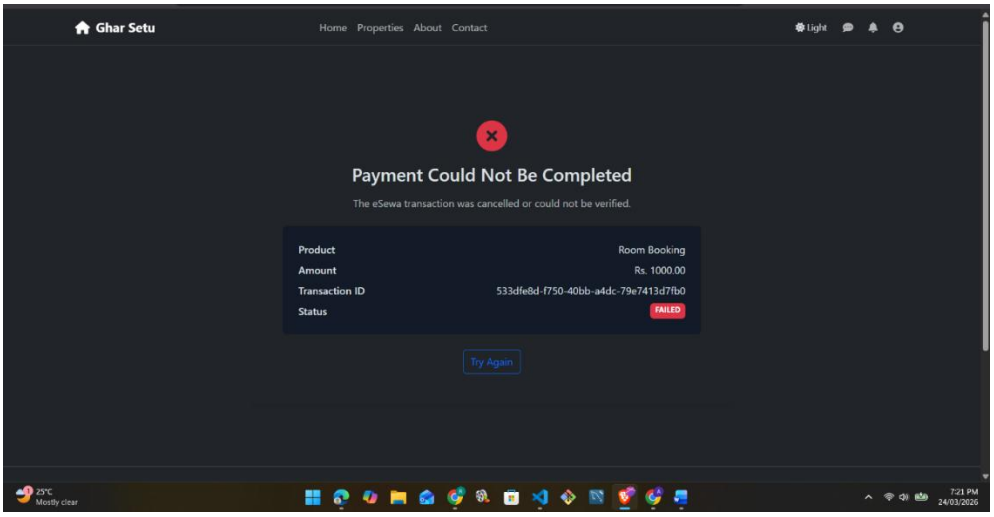

Figure 19: Payment Failure

6.9 Payment Success

Table 16: Test Case 16: Payment Success

Objective	To place the payment and check whether the payment was successful
Input Data	All the process was done for the payment
Expected Output	Payment successful
Actual Result	Your payment completed successfully

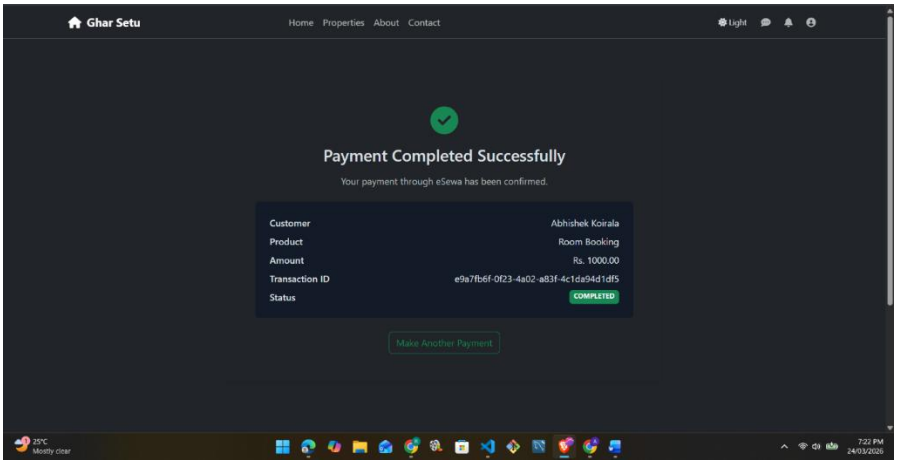

Figure 20: Payment Success

6.10 Chat

Table 17: Test Case 17: Chat Function Successful

Objective	To check whether the chat is working.
Input Data	The message was placed.
Expected Output	Message should be sent and receive a reply.
Actual Result	The chat was functioning successfully.

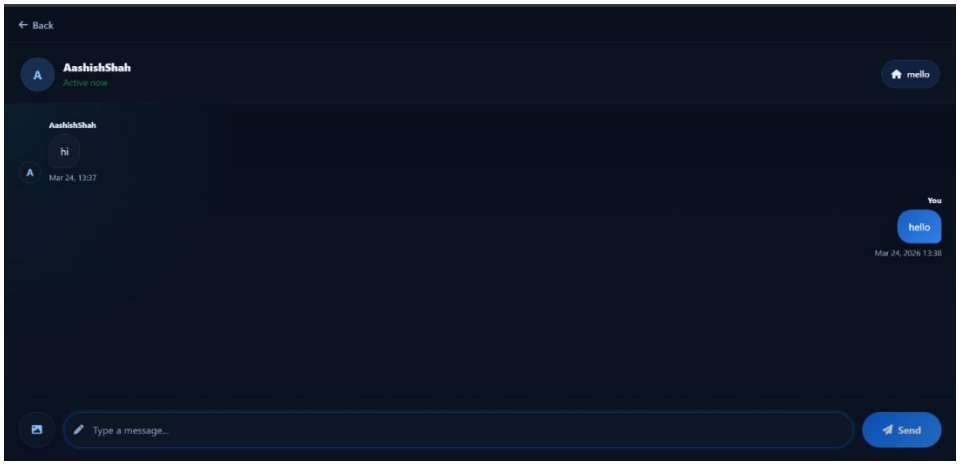

Figure 21: Chat Functions

7 Group Contribution Table

Table 18: Group Contribution Table

Member Name	LMUID	Major Contribution Areas	Key Responsibilities
Aashish Kumar Shah		(Backend Developer - Django)	Django models, views, URLs, Django Channels chat setup, Daphne/Redis config, cloud deployment
Hridya Siwakoti		Frontend Developer (Django Templates)	Designed Using HTML, CSS, JavaScript, Django Templets, created pages, connected Frontend with Django
Anushka Sangroula	24045827	Chat System Developer (Socket.IO Integration)	MySQL schema design, Django ORM queries, migration management, booking and notification backend
Abhishek Koirala		UI/UX Design, Frontend, Documentation	Figma wireframes, CSS custom styling, About and Contact pages, documentation writing

8 Conclusion

The project was implemented to solve the huge inefficiencies and constraints of the current manual process of home rental in the target area, in which the property owners and residents use physical notice boards, personal contacts, and informal means of communication to transact rental deals. The Ghar Setu, which is an online system of management of renting houses was created using Django so that this disjointed process can be replaced through a single and integrated web-based platform.

The project was able to achieve all its set objectives. MySQL database was also normalized and implemented to house all user, property, booking, notification, message, and review information. An entirely responsive front end was developed based on the Bootstrap, HTML, CSS and JavaScript and could be viewed on desktop and mobile browsers. The Django MVT backend was created with secure user/tenant role-based authentication, a full property listing CRUD system with Leaflet map support, a booking request system with automated notifications, a real-time chat system (built on Django Channels and WebSockets), a favourites management system, and a rating and review system. The system was tested with the client on the aspect of user acceptance and found to be able to respond to the practical requirement of a local property management operation.

On the whole, Ghar Setu proves that it is possible to design a technically advanced, feature-rich, and, in other words, locally relevant rental platform within the frame of an educational project with the help of available open-source technologies. The project has given the team a rich practical experience in full-stack Django development, real-time WebSocket development using Django Channels, relational database design using MySQL, interactive map development using Leaflet, Agile project development, professional technical documentation.

8.1.1 Legal Issues

Ghar Setu gathers and archives personal information on registered users such as full names, email addresses, phone numbers, profile photos and personal chat messages. This system should consequently be developed and operated in regard to the legal requirements that are presented by the existing data protection systems. Although there is no stand-alone data protection law in Nepal as comparable to the European GDPR, the Individual Privacy Act 2018 of Nepal provides the statement that people

have a right to the privacy of their personal information, and the organizational bodies gathering it have a duty to treat it appropriately and in accordance with the purpose it was originally obtained.

Based on these values, Ghar Setu gathers only the personal information required in the operation of the rental platform and does not disclose user information to any third parties. The Django administrative interface is only available to relevant administrators and it has a strong administrative password. Any property images and user-generated content on the server are controlled with only one aim that is to fill property listings and user profiles within the platform. The security features that Django has built-in such as CSRF protection and password hashing (PBKDF2) are employed to ensure the integrity and confidentiality of user information. The chat messages between owners and tenants that have transpired after the booking acceptance are stored in MySQL database and can only be viewed by the two parties that have engaged in the booking conversation.

8.1.2 Social Issues

Ghar Setu has good social implications which are meaningful to that which it serves. Digitizing the property discovery and booking process makes the platform expand the property advertisement beyond the immediate physical neighborhood to enable tenants that may be in other parts of the city or country to find and book the available property without a prior physical visit. This is especially applicable in Nepal, where people are moving to urban areas in huge numbers, which results in an enduring need of renting accommodation in the people who have not yet gained access to a local social network.

The site further facilitates impartiality and openness in the renting market through establishing a structured, documented history of everything requested in booking, choices, and interactions. This limits the possibility of disputes after oral agreements having been formed and provides all owners as well as tenants a clear and equal record of their interactions. The rating and review system also encourage accountability as tenants get to contribute their experiences of particular properties and landlords and in the long run, it creates a layer of trust that is community-sourced hence beneficial to future users.

It is, however, admitted that the advantages of Ghar Setu depend on whether the users

have access to an internet-connected device and a certain degree of digital literacy. Some of the targeted beneficiaries of the platform might not be able to engage in the platform in locations which cannot access smartphone or the internet. To minimize such barriers to access, future development should consider such features as an embedded SMS notification fallback or a multilingual interface in Nepali.

8.1.3 Ethical Issues

Several ethical concerns were used to design and develop Ghar Setu. The system considers user privacy as an imperative right. Personal information, contacts and private chat messages are all stored in a safe location and are secured by role-based access controls and are never made available to any other user other than the one directly engaged in the transaction in question. The platform neither records the behavior of users to advertise them nor sell their data to a third party in any ways.

The rating and review system were created in such a way that is fair. Registered users that have ever communicated with a property on the platform can leave a review and there is no system to allow property owners to block or delete negative reviews. This makes sure that the review system remains an accurate user experience and not a positive feedback that was edited. The booking acceptance and rejection process should be open to both the owner and the tenant, and the result of all the booking choices should be declared to both the owners and the tenants through the system and the system must record the reason why the booking is rejected in the system to be reviewed by the administration in case of a dispute.

One of the ethical considerations that led to the selection of the Leaflet and OpenStreetMap integration is that the system is completely based on open-source and free mapping data, unlike commercial mapping services, which can potentially place limitations on the use of data or have geographical bias. This also makes sure that the platform is free to run without commercial costs of API that would jeopardize its sustainability in the long run as a community resource that is easily accessible.

8.1.4 Future work

Despite the success of Ghar Setu in terms of delivering its main goals within the academic project deadline, it is possible to mention several improvements that should be made in the future. First, an indigenous mobile Android/iOS app would largely enhance the level of accessibility among users who largely use smartphones as the type of device to access the internet in Nepal since it is the predominating device in the country. Second, by including a digital payment gateway (eSewa, Khalti, or bank transfer confirmation) tenants would be able to pay the first rental deposit and monthly rent directly on the platform, establishing an entirely digital rental experience. Third, the interface option of the Nepali language would significantly reduce the barrier to entry by the user who can speak his or her language more comfortably. Fourth, property discovery should be enhanced with the introduction of a more advanced search module that has more filter options, including number of rooms, floor level, furnishing status, and distance to landmarks, including schools or bus stops. Fifth, a dashboard of landlord analytics with views on listings, statistics on booking requests, and occupancy rates would provide property owners with useful information about the performance of their listing. Sixth, an artificial intelligence-based property recommendation engine that would recommend the relevant listing to the tenants depending on their browsing history and favourites would improve user engagement and satisfaction considerably. Seventh, the platform would need a formal lease agreement generation module just like the case with Home Rental Management to complete the rental lifecycle management of the platform.

9 References

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magicbricks. (2026) *Properties for the Global Indian!* [Online]. Available from: https://www.magicbricks.com/ [Accessed 21 February 2025].

nepalpropertybazaar. (2026) *Leading Real Estate in Nepal* [Online]. Available from: https://nepalpropertybazaar.com/ [Accessed 21 March 2026].

10 Appendix

10.1 Appendix A: Interview Question to the client.

The questions that were applied during the initial client requirement gathering meeting with the local property management intermediary included the following:

- How many properties do you currently manage on behalf of individual owners, and how is this information currently organized?
- How do you advertise the newly offered rental house and how long does it normally require to get a good tenant?
- How do you now receive inquiries of prospective tenants seeking a property; what are the issues that you encounter in the process?
- What is the current system of communicating rental decisions to tenants and the agreement.
- Have you had any conflicts with the owners and tenants related to miscommunication or unrecorded contract?
- Does your organization currently use any digital tools or platforms in any aspect of the rental process?
- Which attributes do you believe would be the most significant in a digital platform that will take the place of your current process?
- Do you or your customers feel at ease using a map-based interface to mark the location of properties?
- Which are your most frequently used devices with your clients to gain access to the internet?

10.2 Appendix B: User Survey

A short user survey was given to 25 people of Itahari region who had recently rented a house or attempted to rent a house. The most important findings were noted as the following:

- 88 percent of the respondents indicated that they had used word-of-mouth as a tool to get available rental properties.
- Three-quarters of them (76 percent) indicated that they had toured over three properties in person before locating one that fit their needs.
- According to the survey, 92 percent said that they would first like to search online and look at any available properties, before they could devote their time to a physical visit.
- A majority of 84% indicated that they would feel more reassured by having a system of booking request that provided them with a formal response by the owner as opposed to the informal process used in the current system.
- 79% indicated they would be interested in having access to a landlord via the in-app chat tool once they had made their booking.
- 91% of respondents indicated that they would appreciate having honest reviews and ratings by past tenants in their decisions when renting property.

11 Developed Project (Screenshots / System Demo / Wireframe)

11.1 Figma Designs

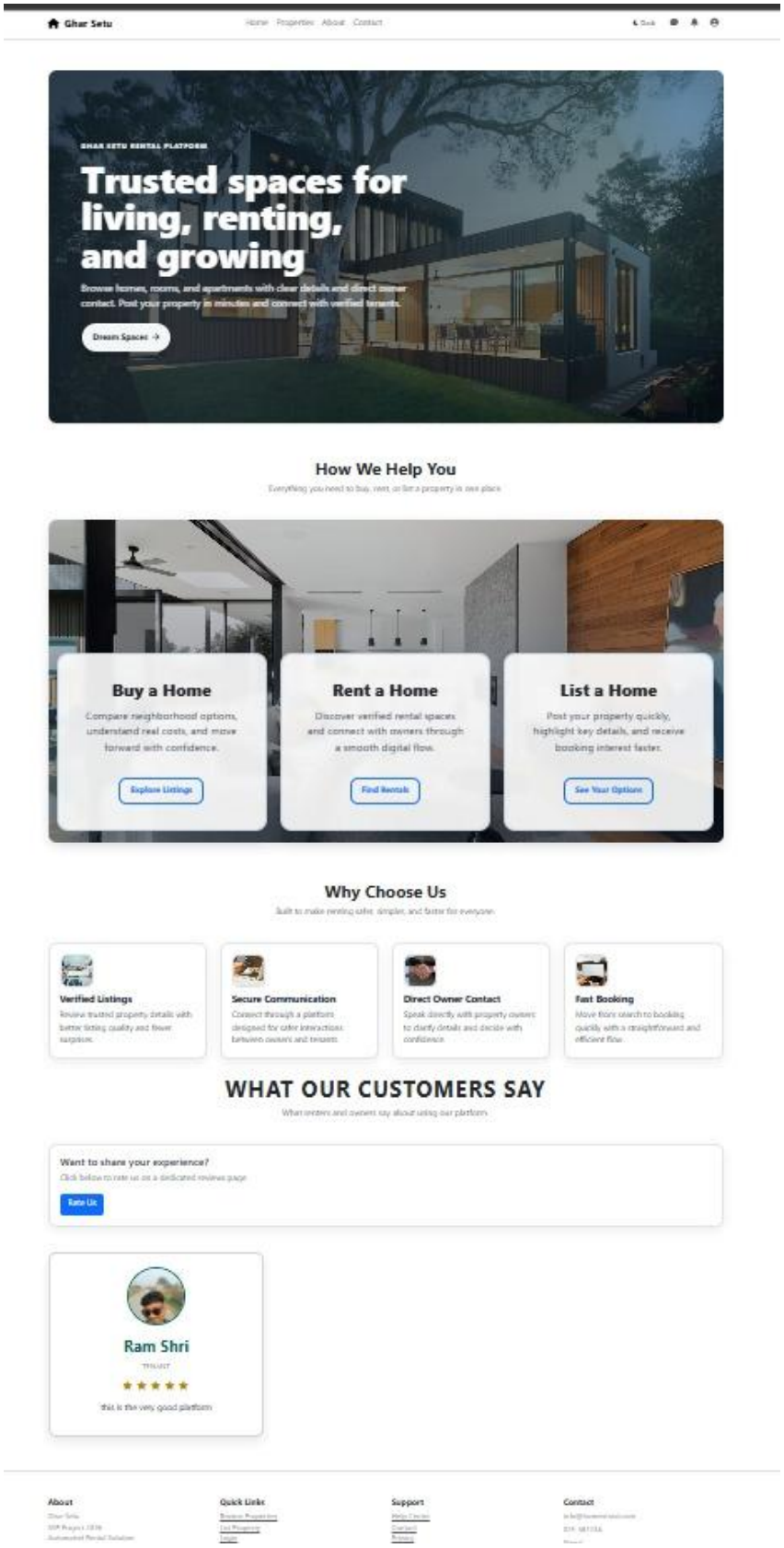

Figure 22: Home page Design

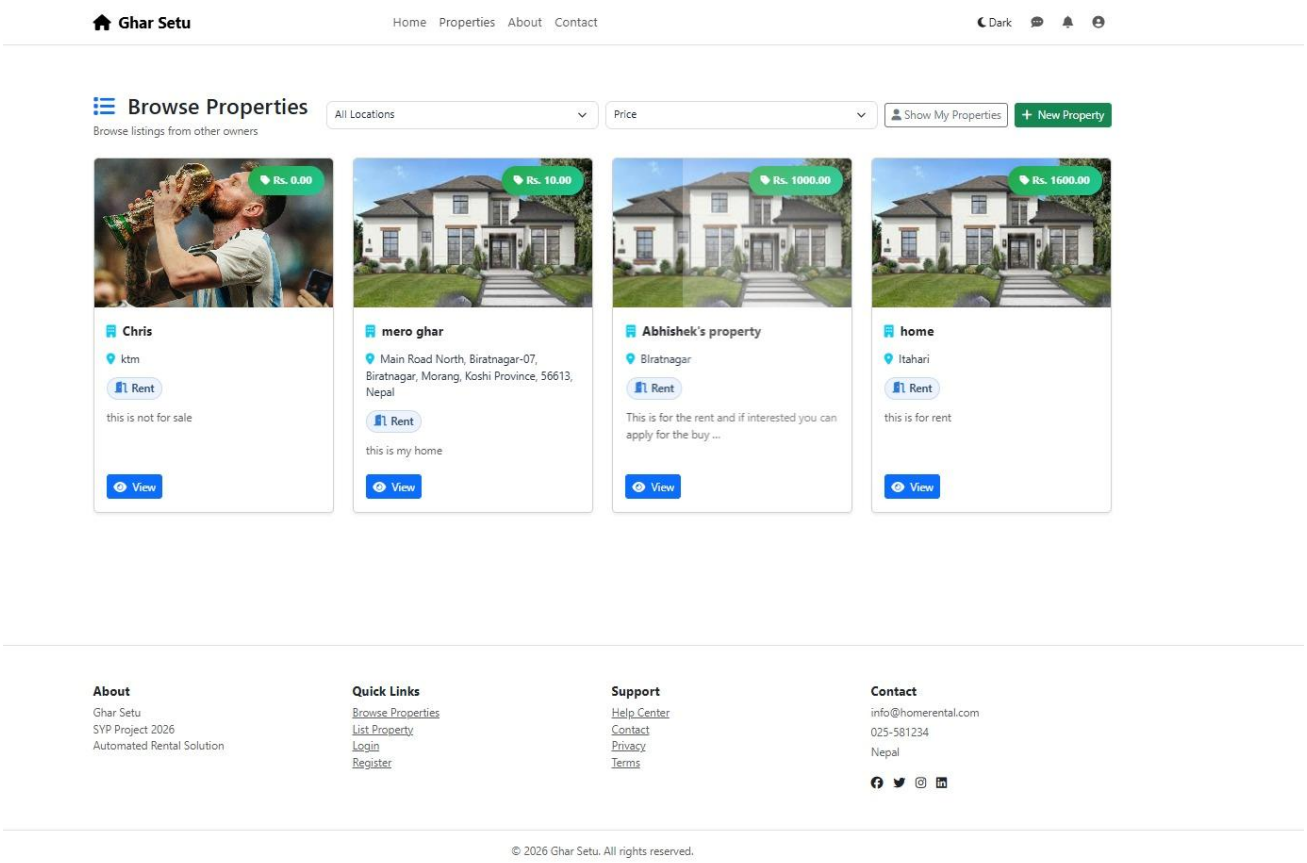

Figure 23: Properties Page Design

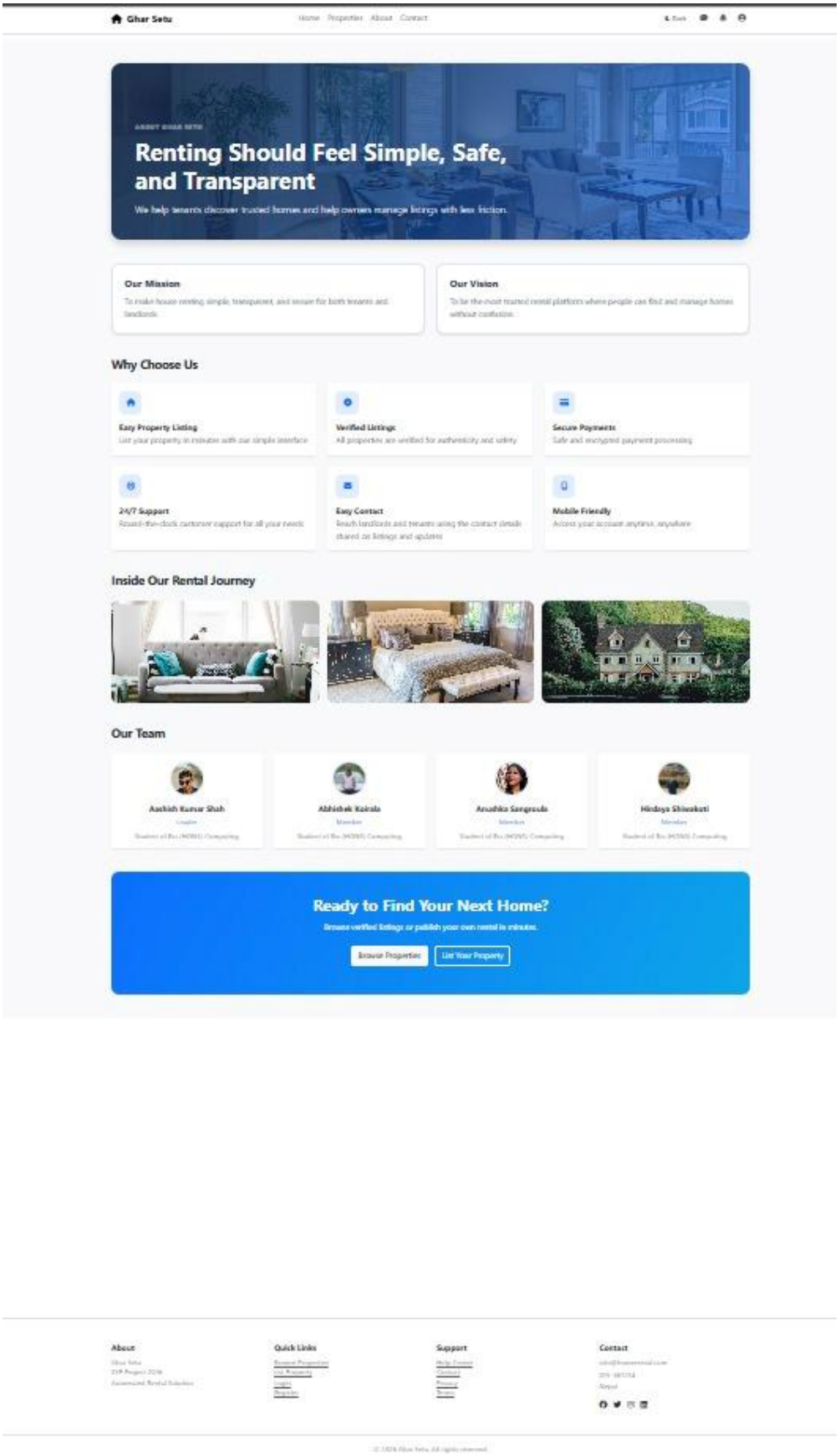

Figure 24: About Us Page Design

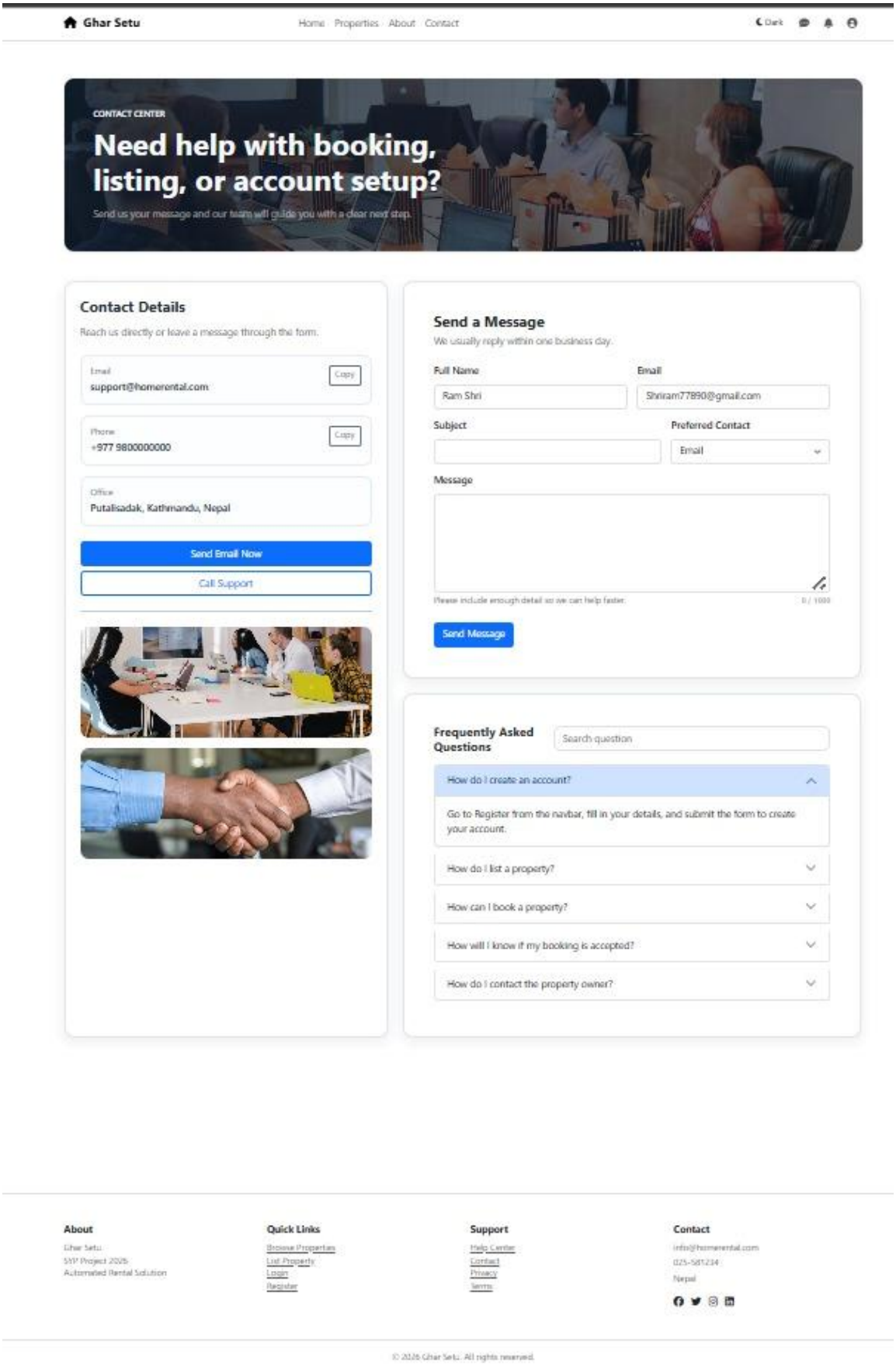

Figure 25: Contact Us Page Design

11.2 Actual Website Pages

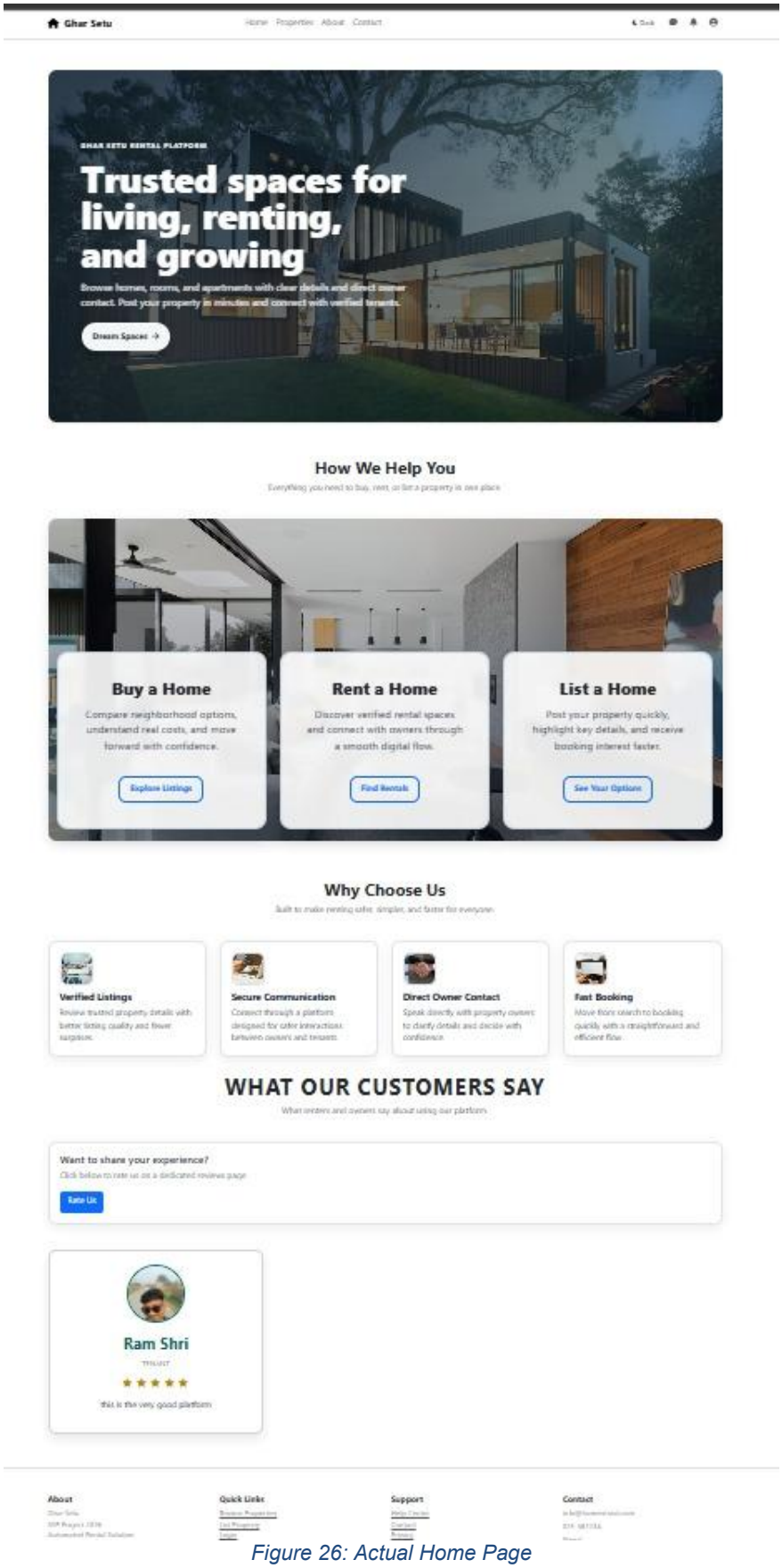

Figure 26: Actual Home Page

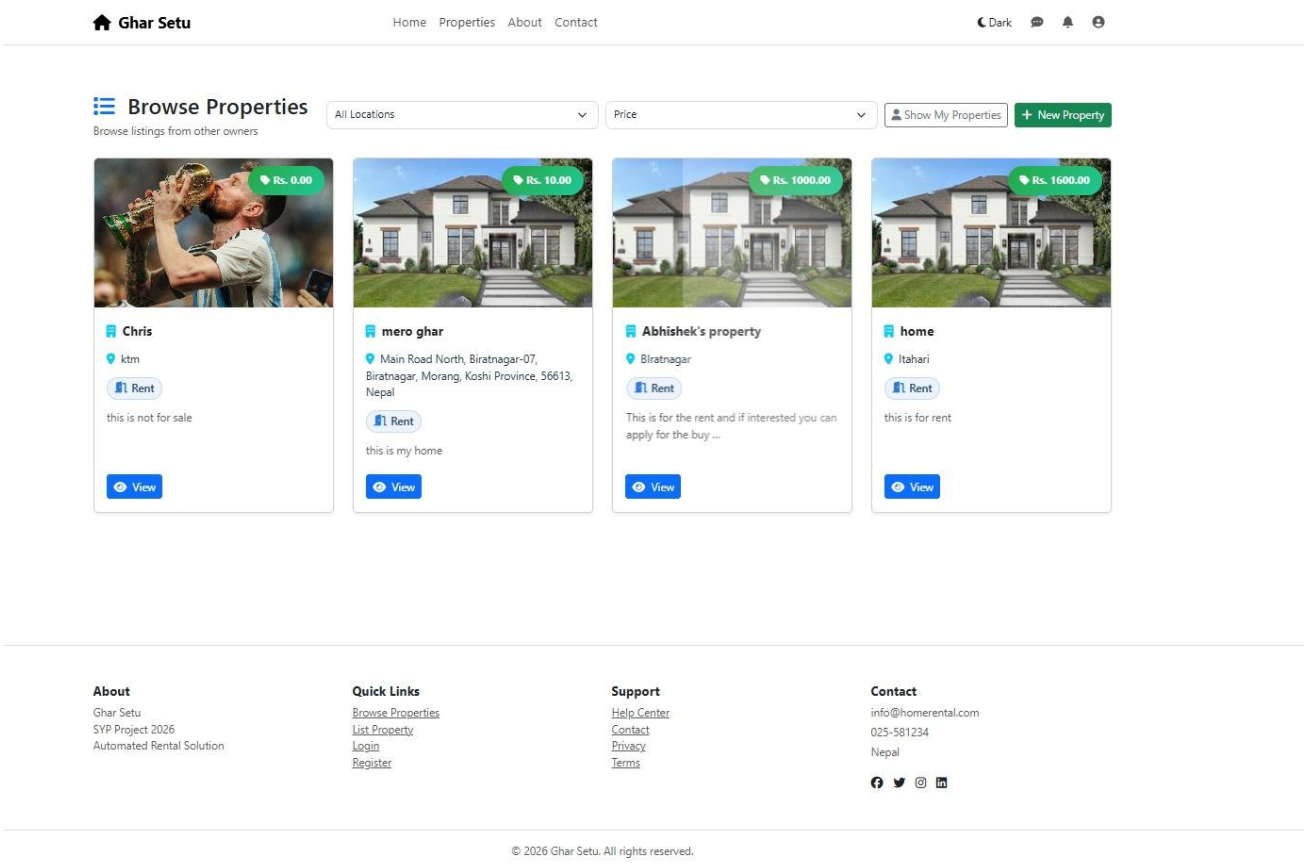

Figure 27: Actual Properties Page

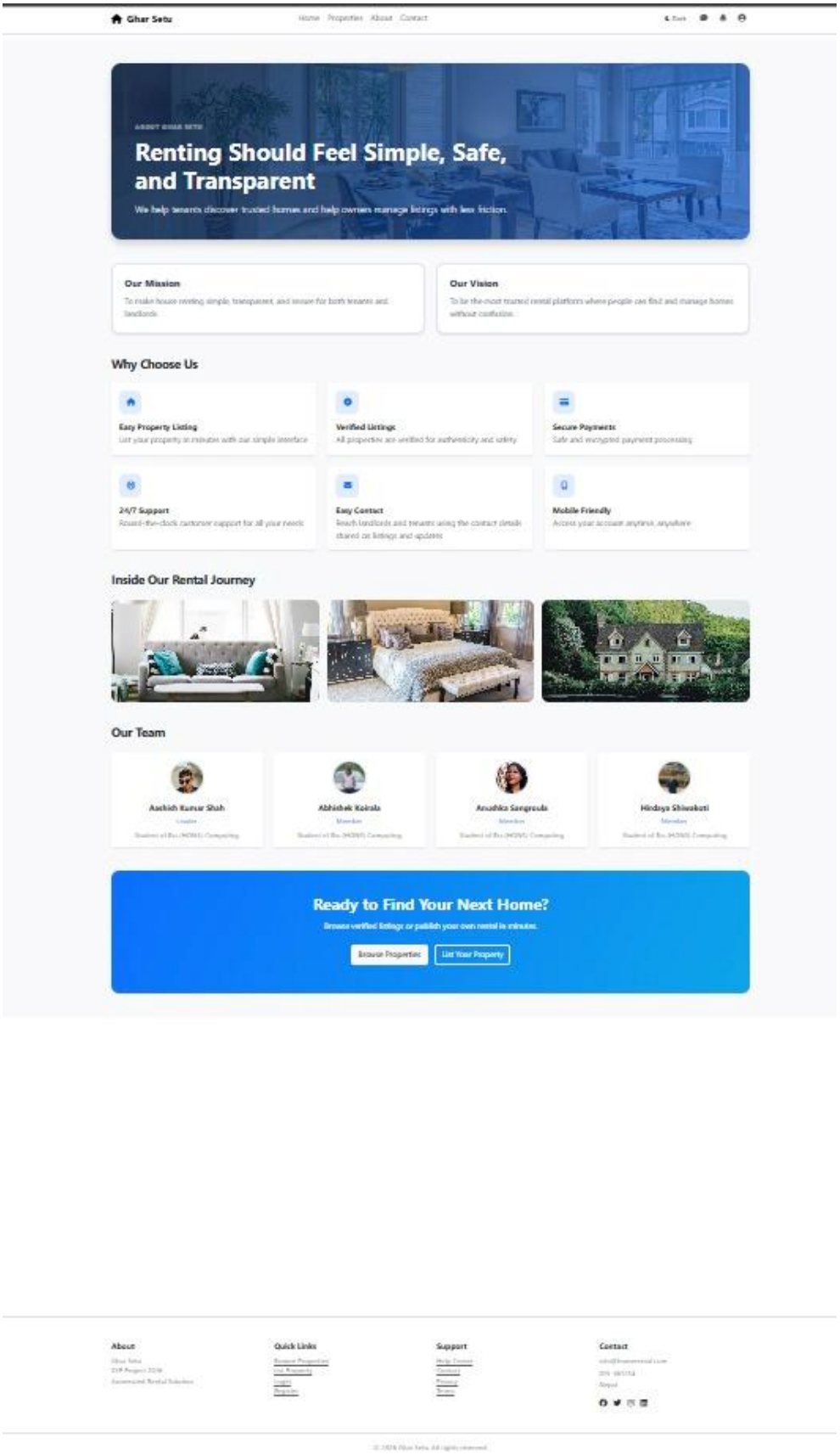

Figure 28: Actual About Us Page

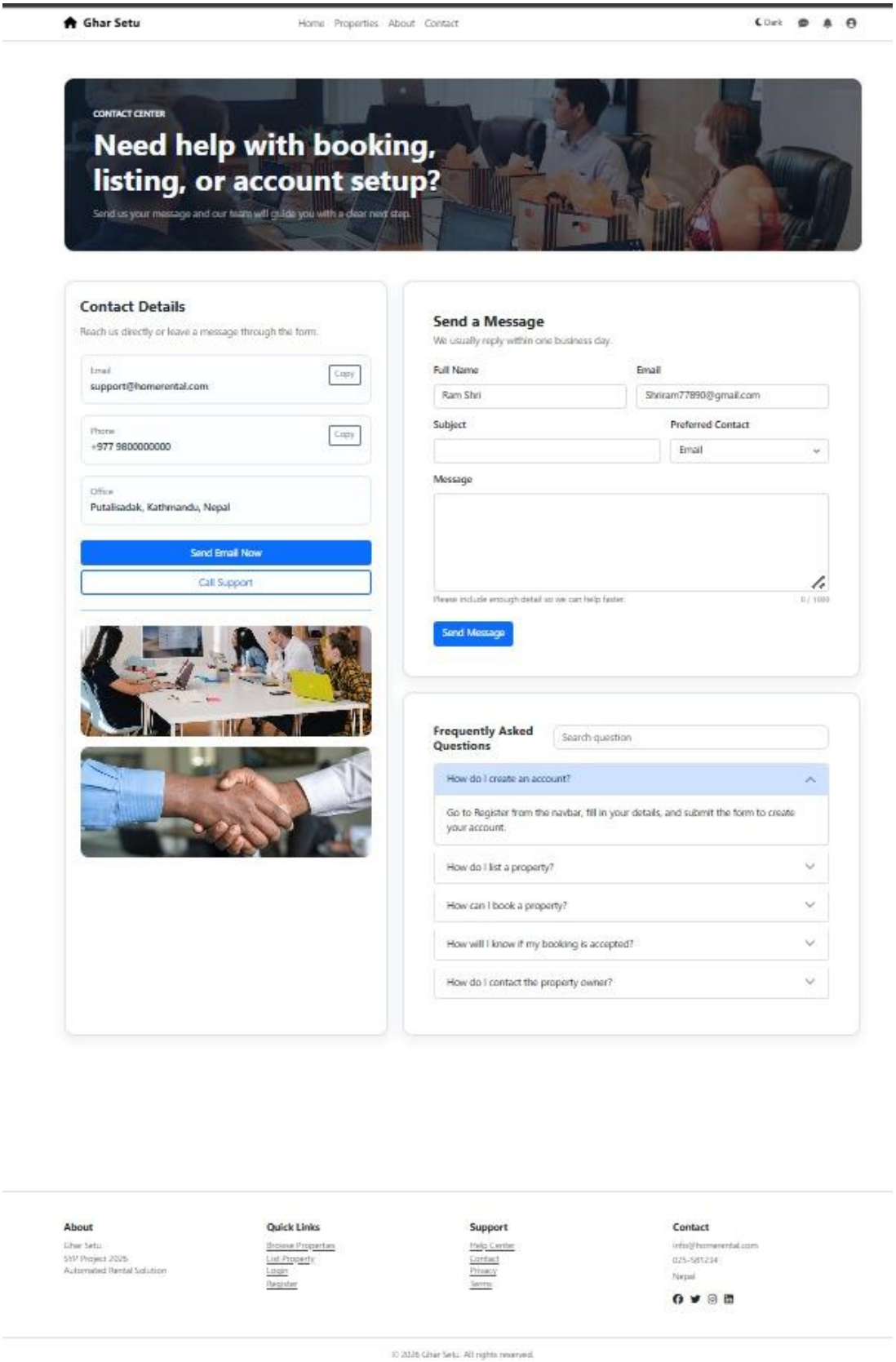

Figure 29: Actual Contact Us Page